The Local-global Equivalence on General Networks: Theory*

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Abstract

Strategy-proofness is an incentive property requiring that no (preference) type gain by reporting any other type in a prespecified domain of admissible types. The "local-global equivalence (LGE)" refers to the equivalence of strategy-proofness and "local" strategy-proofness, which requires immunity to misrepresentation assuming that an agent is constrained to report "local" types relative to his true type. Generalizing an existing framework, we allow the notion of localness to be arbitrary and directed and hence manipulation opportunities to be asymmetric across types. We identify two conditions, each of which characterizes the networks satisfying LGE: (i) strong connectedness, a refined notion of connectedness; and (ii) Property DL, a directed-network adaptation of Property L by Kumar et al. (2021a). This characterization also helps reveal conditions for the random version of LGE (the equivalence of strategy-proofness and local strategy-proofness for random rules). We discover a necessary condition called hyper-connectedness and a sufficient condition called Property

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ULLO for random LGE. Our conditions are more general than earlier ones in the literature and are satisfied by several well-known preference domains.

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1 Introduction

1.1 Motivation

In the theory of mechanism design, incentive compatibility is a desideratum of chief interest. It requires that no individual ever gain by misrepresenting his private information, or "type". Typically, an agent can claim to be of any type in a given type space. An apparently much weaker property, called local incentive compatibility, only requires that "local" deviations from one's true type should never benefit him. In a wide range of contexts, it has been shown that local incentive compatibility turns out to be equivalent to the usual notion of incentive compatibility. Understanding this "local-global equivalence (LGE)" is our goal. In our setting, an agent's type is his ordinal preferences over alternatives and he is constrained to report a type within a prespecified set, as captured by an arbitrary network defined over types. We characterize networks satisfying LGE for deterministic mechanisms. We also provide a necessary condition and a sufficient condition for LGE when randomization is allowed.

A standard assumption in the mechanism design literature is that individuals may misrepresent their types and they have considerable freedom in doing so. But there are several reasons to revisit the assumption. The first concerns feasibility of lies. In some contexts, an agent may have to produce hard evidence to substantiate his report and only a limited form of manipulation, such as withholding information, is available to him (Grossman, 1981; Milgrom, 1981; Green and Laffont, 1986; Che and Gale, 2000; Sher and Vohra, 2015). Cred-

ibility may also restrict the set of plausible options for misrepresentation. If some information has been revealed about his type in a prior stage, it becomes a constraint on his lies: credible lies should align with that information.

Second, the cost of manipulation can be a deterrent and steer manipulation in particular directions. The very act of lying is costly in some settings. For example, an agent may have to fabricate a financial report to convince a principal. Then the agent will find it optimal to manipulate his message only a little (Maggi and Rodriguez-Clare, 1995). Further, lying, if detected, sometimes has material consequences and more, as in the case of tax evasion (Allingham and Sandmo, 1972). Smaller lies are harder to catch, which lowers their expected cost, thereby making them first candidates for misrepresentation.

Third, a growing body of empirical and experimental literature shows that individuals have intrinsic preferences for honesty and promise-keeping even when they come at a personal cost. Tax compliance rates often exceed the levels predicted solely by negative incentives such as penalty. Andreoni et al. (1998) and Torgler (2002) attribute it to moral and social codes that guide tax reporting decisions. In field and lab experiments too, subjects choose to incur a cost and be honest (Nagin et al., 2002; Gneezy, 2005; Hurkens and Kartik, 2009; Charness and Dufwenberg, 2006; Sânchez-Pagés and Vorsatz, 2007; Vanberg, 2008; Mazar et al., 2008; Fischbacher and Föllmi-Heusi, 2013). Indeed, some stylized patterns in the observed data are supported as an equilibrium play if (perceived) lying aversion[1] is incorporated into a theoretical framework (Kartik, 2009; Dufwenberg and Dufwenberg, 2018).

Once we accept the proposition that individuals manipulate but not necessarily maximally, some interesting questions emerge. Suppose that types are ordinal preferences over a finite set of alternatives. For each type, there is a set of "local" types that the agent can credibly announce. This set may consist of the "smallest lies" relative to the truth, namely those obtained by switching a pair of alternatives adjacent in rankings (Carroll, 2012; Sato, 2013a). In general,

[1]Perceived lying aversion, distinct from lying aversion, refers to the aversion to being perceived to be lying when one is observed by an audience (Dufwenberg and Dufwenberg, 2018).

the specification of local types is represented by a network defined over types. Each network gives rise to a notion of local incentive compatibility. Then one can investigate properties of the network guaranteeing that the associated notion of local incentive compatibility implies full-blown incentive compatibility, i.e., whether and when LGE holds. The importance of LGE lies also in the unwieldy complexity of verifying a complete set of incentive constraints and the often negative consequences it entails, such as the Gibbard-Satterthwaite Theorem (Gibbard, 1973; Satterthwaite, 1975). Can we determine whether a subset of local incentive constraints ensures incentive compatibility? Does replacing incentive compatibility with local incentive compatibility expand the menu of available mechanisms to more palatable ones? Can we provide convenient tests for LGE that can be applied to existing weaker incentive properties to explore their implications? Finally, if randomization over alternatives is permitted, how can we generalize sufficient or necessary conditions for LGE? We answer these questions.

1.2 Our Model

To ease exposition, we consider a single-agent setting where a type is the agent's ordinal preferences over finitely many alternatives.[2] A domain is a set of admissible types. A rule (mechanism) takes the agent's report (truthful or not) of his type as input and chooses an alternative. *Strategy-proofness* requires that no type should benefit from reporting any other type in the domain. In practice, however, a type may face a constraint when manipulating. The constraint specifies a set of local deviations the type can report. It is captured by a network constructed as follows: there is a directed edge from type P to type P' in the domain if P can claim to be P'. Edges are *directed*: even though P is able to announce P', P' may not be able to announce P. An environment (network) is a pair $\langle \mathcal{D}, \mathcal{E} \rangle$, where $\mathcal{D}$ is a domain treated as a set of vertices (types) and $\mathcal{E}$ is a set

[2]The single-agent setting is without loss of generality because strategy-proofness involves comparing the outcomes an agent obtains by manipulating, taking the other agents' reports as given.

of edges. Figure 2 below is an example. Given an environment, *local strategy-proofness*, a weakening of strategy-proofness, requires that for all types P and P' in the domain with an edge from P to P', type P should not benefit from reporting P'. An environment is said to satisfy *LGE* if every locally strategy-proof rule defined on the environment is, in fact, strategy-proof.

Our model extends the general framework on LGE due to Kumar et al. (2021a) by allowing asymmetry of misrepresentation among types. This yields directed networks, as opposed to undirected ones in Kumar et al. (2021a). Our objective is to accommodate cases where announcement constraints need not be symmetric across types. A type would be constrained in manipulation if his claims should be supported by hard evidence such as certificates or some of his private information had been revealed in an earlier stage; e.g., see Green and Laffont (1986), Milgrom and Roberts (1986), Okuno-Fujiwara et al. (1990), and Che and Gale (2000) (the literature review in Section 1.4 provides more details). The evidential requirement or the revealed information is likely to vary across types, giving rise to different manipulation constraints for different types. As a more concrete example, suppose that there are three alternatives, $\{a, b, c\}$, and the domain includes all six types (strict preference relations). Assume that type abc (for which a is the best and b is the second best) has already revealed his true preferences over one pair of alternatives, say, $a \succ c$ ("a is preferred to c"). To be consistent with the revealed information, type abc's announcement should be in $\{abc, acb, bac\}$. On the other hand, if type bac has revealed $b \succ a$, he should choose from $\{bac, bca, cba\}$. Then misrepresentation is asymmetric across types because type abc can claim to be type bac but not the other way around.

Even when the above scenario is of lesser concern, directed networks are needed if one seeks to identify a minimal set of local incentive constraints that is sufficient for strategy-proofness. By definition, an edge in undirected networks represents bidirectional local incentive constraints. Restricting attention to undirected networks, we cannot determine whether some local constraints captured by undirected edges are indeed redundant in ensuring strategy-proofness.

Networks that are minimal in the sense of having a smallest set of edges among all networks satisfying LGE will emerge only if we adopt directed networks and test the necessity of each direction of each edge.

Directed networks are also useful when we study non-manipulability properties associated with heuristic strategic behavior. If agents do not always lie to the fullest extent, it is conceivable that they will resort to some simple operations to come up with misrepresentations. Altuntaş et al. (2023) consider a setting where agents only drop a single alternative to the bottom of the preference list or push it up to the top.[3] Their incentive properties, called *drop strategy-proofness* and *push-up strategy-proofness*, require immunity to such manipulations. Since dropping and pushing up induce asymmetric misrepresentation among types, their connection to strategy-proofness can easily be investigated with directed networks.

1.3 Main Results

Our first result is a characterization of the environments satisfying LGE. We provide two conditions on environments, "strong connectedness" and the "directed lower-contour-set no-restoration property" (simply, Property DL), each of which is equivalent to LGE (Theorem 1). To illustrate our conditions, let us begin by recalling a simple necessary condition for LGE, identified by Sato (2013a) for "adjacency environments". In an adjacency environment, there is an edge from one type to another if their preferences differ only over a single pair of alternatives (such alternatives are adjacent in rankings; Kemeny, 1959).[4] Say that an environment is *connected* if for all types P and P' in the domain, there is a path, i.e., a sequence of edges, from P to P'. For example, for the environment in Figure 2 below, (abc, bca, cba) is a path from type abc to type cba. Each

[3]In a model of object allocation, Altuntaş et al. (2023) propose and characterize the generalized top trading cycles rule. They also prove some impossibility theorems. Their results involve properties weaker than strategy-proofness, such as drop strategy-proofness, push-up strategy-proofness, and upper invariance.

[4]This is a special case of the environments studied by Kumar et al. (2021a) and hence of our environments.

environment satisfying LGE turns out to be connected (Sato, 2013a; Kumar et al., 2021a). If an environment is not connected, so that for some P and P', there is no way of reaching P' from P, then one can construct a locally strategy-proof rule for which P is better off reporting P', violating strategy-proofness. Our strong connectedness expands the spirit underlying connectedness and applies it to various components of the given environment, as follows.

Given an environment $\langle \mathcal{D}, \mathcal{E} \rangle$, we decompose it into several "subenvironments", each of which consists of types that have common preferences over some alternatives. Subenvironments are indexed by (a, B), where B is a subset of alternatives and a is an alternative in B. The subenvironment associated with (a, B) has (i) the subdomain consisting of the types in $\mathcal{D}$ that most prefer a in B; and (ii) the set of edges restricted to the latter subdomain. A subenvironment can be viewed as an environment on its own. *Strong connectedness* requires that each subenvironment be connected (i.e., for all P, P' in the subdomain, there should be a path from P to P' that does not leave the subenvironment). It is intuitive that LGE implies strong connectedness—just as LGE implies connectedness. Rather unexpectedly, the converse also holds: strong connectedness implies LGE.

Our second condition, *Property DL*, is a directed-network version of Property L, a condition identified by Kumar et al. (2021a). The focus of the latter paper is on "symmetry" environments, in which edges are undirected, so that if P can report P', then P' can also report P. For symmetry environments, Property L requires that for all types P and P' and each alternative a, there is a path between P and P' such that for each b in the lower contour set of P at a, the path is without $\{a, b\}$-restoration (i.e., along the path, the relative preferences over a and b should be reversed at most once). Kumar et al. (2021a) show that Property L characterizes LGE for symmetry environments. In our environments of asymmetric misrepresentation, we adapt Property L by taking account of the directed nature of edges. We show that the resulting property, which we call Property DL, is necessary and sufficient for LGE.

The equivalence of LGE, strong connectedness, and Property DL casts new light on LGE in several respects. Clearly, it generalizes the characterization of LGE by Kumar et al. (2021a) to environments allowing asymmetric manipulation. More significant, however, is that LGE is characterized by strong connectedness. As our examples below indicate, strong connectedness is a convenient test for LGE based on visual inspection. Further, it is rooted more in connectedness, unlike the existing conditions, Property L (Kumar et al., 2021a) and the no-restoration property (Sato, 2013a), that build on the concept of no-restoration. Therefore, our result shows how one can refine notions of connectedness and no-restoration to capture LGE equivalently.

Next, we turn to LGE for random rules. Say that an environment satisfies *random LGE* (or *RLGE*) if each locally strategy-proof random rule is strategy-proof.[5] RLGE is a stronger requirement on environments than LGE and turns out to be much harder to characterize. We provide a necessary condition and a sufficient condition for RLGE. To introduce our necessary condition, we further refine strong connectedness, decomposing an environment into a larger collection of subenvironments than strong connectedness does. Now subenvironments are indexed by (C, B), where B and C are subsets of alternatives with $C \subsetneq B$. The subenvironment associated with (C, B) collects (i) the types that prefer each alternative in C to each alternative in $B \backslash C$ (so that when the types are restricted to B, C is an upper contour set); and (ii) the edges restricted to those types. Say that an environment $\langle \mathcal{D}, \mathcal{E} \rangle$ is *hyper-connected* if all such subenvironments are connected. Clearly, hyper-connectedness strengthens strong connectedness because in (C, B), C can be a singleton. We show that each environment satisfying RLGE is hyper-connected (Theorem 2).

While hyper-connectedness is also a tractable condition one can visually verify, it is not sufficient for RLGE in general environments and neither is the

[5]Since the agent is equipped only with ordinal preferences over lotteries, he applies first-order stochastic dominance to compare lotteries selected by random rules (Gibbard, 1977; Bogomolnaia and Moulin, 2001). Strategy-proofness and its local version is defined using the preferences so obtained.

no-restoration property of Sato (2013a).[6] To recover sufficiency, we exploit the spirit of no-restoration inherent to Property DL. In rough terms, Property DL requires no-restoration between any given alternative and its lower contour set. As we shift to random rules, probabilities for upper contour sets come into play since preferences based on first-order stochastic dominance are used to compare them. Therefore, an RLGE counterpart of Property DL should require no-restoration between any upper contour set and its complement (i.e., its lower contour set). However, this is not enough. Another condition should be imposed on the required path to govern the manner in which alternatives in the chosen upper contour set become less desirable along the path. This condition, together with no-restoration between upper and lower contour sets, gives us what we call the *upper-lower no-restoration property with limited overtaking*, or simply *Property ULLO*. We show that if an environment satisfies Property ULLO, it satisfies RLGE (Theorem 3). Our sufficient condition is met by a wide range of environments that proved fruitful in the literature, including the universal, single-peaked, single-dipped, single-crossing, hybrid, and lexicographically separable domains.

The question of RLGE remains scantily explored in the literature. To our knowledge, no necessary condition for RLGE has been identified so far[7]; only one sufficient condition is available (Cho, 2016) and its applicability is limited to adjacency environments. When restricted to adjacency environments, Property ULLO is substantially weaker than the sufficient condition by Cho (2016). More importantly, hyper-connectedness and Property ULLO are the first criteria for checking RLGE of general environments.

The usefulness of our results is seen more clearly when our conditions—strong connectedness, hyper-connectedness, and Property ULLO—are put to work. In our companion paper, Cho and Park (2025), we apply our main results

[6] In adjacency environments, however, the no-restoration property does imply RLGE (Cho, 2016).

[7] To be strict, since RLGE implies LGE, conditions necessary for LGE, including those identified by Sato (2013a), are also necessary for RLGE. However, no condition appears to be necessary just for RLGE.

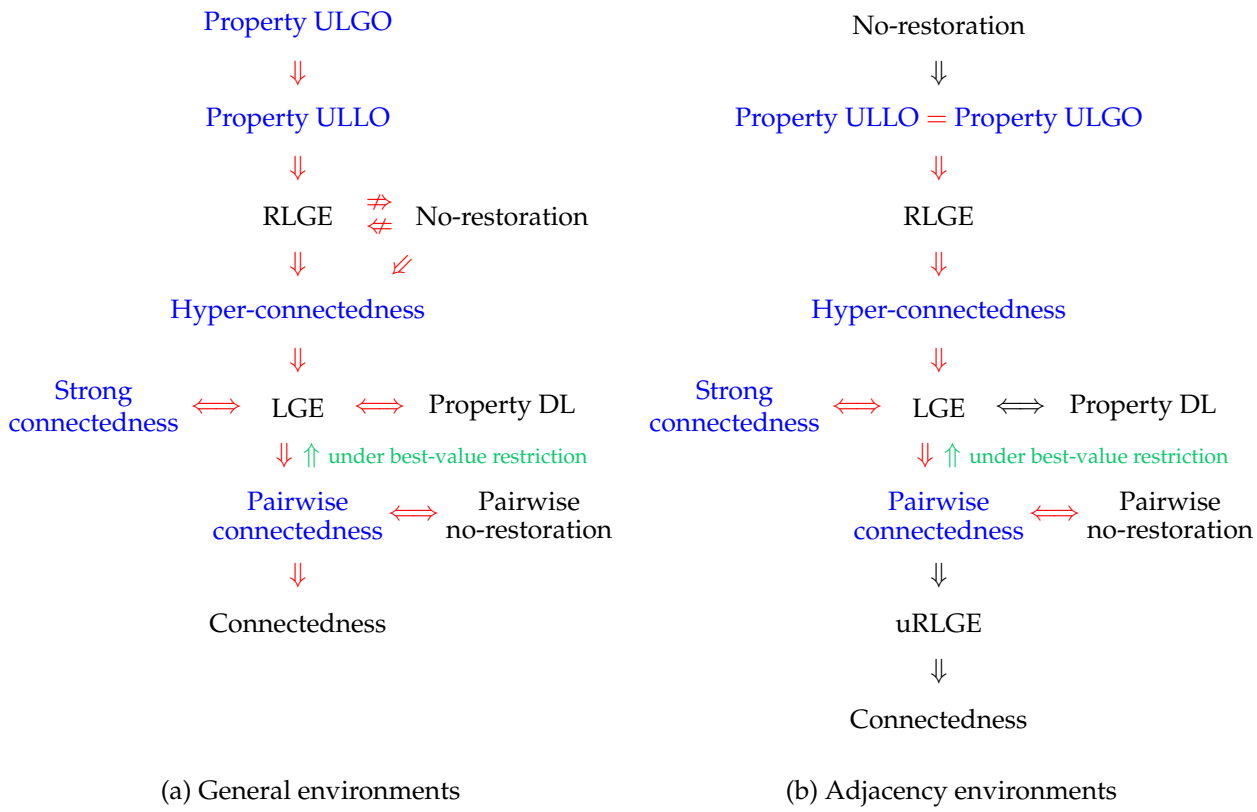

Figure 1: The logical relationship among (R)LGE and properties of environments (the properties we identify are in **blue** and the implications we prove are in **red** and **green**)

to questions that the previous work on LGE (Sato, 2013a; Cho, 2016; Kumar et al., 2021a,b) cannot address as they only consider symmetry environments or their sufficient conditions for RLGE are too restrictive. Among these applications are the issues mentioned in Section 1.2, of constructing LGE-minimal environments and establishing the connection between strategy-proofness and heuristic incentive properties (drop and push-up strategy-proofness and their variants). The applications illustrate tractability of our approach and results.

Our investigation produces quite a few properties of environments. To provide a better view of our contribution relative to existing properties, we summarize the logical relationship among (R)LGE and various properties in Figure 1. Panel (a) shows the relationship for general environments (including asymmetry ones). The relationship simplifies when we restrict attention to adjacency environments, as shown in panel (b).

1.4 Related Literature

Carroll (2012) and Sato (2013a) are seminal contributions to the LGE literature. Carroll (2012) provides a unified approach to LGE that can be applied to a broad range of mechanism design problems where the type space is discrete or continuous and the mechanism is deterministic or probabilistic. On the other hand, Sato (2013a) studies a deterministic social choice model. He shows that connectedness is necessary for LGE and that the no-restoration property (and a slightly weaker property than this) is sufficient.[8] The two papers identify the universal domain, the single-peaked domain, a restricted single-crossing domain as examples satisfying LGE. In the realm where these papers overlap, the definition of local manipulation is based on Kemeny-adjacency (Kemeny, 1959).

In an important departure from the Kemeny-adjacency approach, Kumar et al. (2021a) provide a framework that can accommodate an arbitrary notion of localness. They characterize LGE by Property L and show, with the aid of the characterization, that the separable domain and the multi-dimensional single-peaked domain satisfy LGE. As mentioned above, our finding that LGE, strong connectedness, and Property DL are equivalent extends the theorem by Kumar et al. (2021a) to environments permitting asymmetric misrepresentation. It reveals a new connectedness structure that LGE entails. Kumar et al. (2023) also study the question of LGE for directed networks.[9] To use our terminology, they prove the equivalence of LGE and Property DL in the context of deterministic rules. The latter equivalence follows from our result.

Concerning RLGE, Cho (2016) shows that for adjacency environments, the no-restoration property (Sato, 2013a) suffices for RLGE. Established for general environments, our necessity and sufficiency results, involving hyper-connectedness and Property ULLO respectively, are of wider applicability than Cho's (2016). Further, Property ULLO significantly weakens the no-restoration property. One

[8] Sato (2013a) and most subsequent papers study domains consisting of strict preference relations only. In a companion paper, Sato (2013b) allows preference relations to exhibit indifferences and finds a sufficient condition for LGE.

[9] After completing this paper, we learned that Kumar et al. (2023) independently investigate LGE for directed networks.

way of circumventing the difficulty associated with analyzing RLGE is to impose a mild requirement, such as unanimity, together with local strategy-proofness and see if it takes us closer to strategy-proofness. An environment is said to satisfy uRLGE if every locally strategy-proof and unanimous random rule is strategy-proof. Kumar et al. (2021b) show that for adjacency environments, the pairwise no-restoration property—a weakening of the no-restoration property—implies uRLGE. By Sato (2013a), we already know that the pairwise no-restoration property is necessary for LGE. Thus, the result by Kumar et al. (2021b) suggests that adding unanimity weakens RLGE to the extent that uRLGE is weaker than LGE.[10]

Though not directly related, several papers in other strands of literature address sufficiency of local incentive compatibility. Our notion of incentive compatibility is dominant strategy incentive compatibility (i.e., strategy-proofness). In investigating ordinal Bayesian incentive compatibility (d'Aspremont and Peleg, 1988), a different notion, some papers derive implications of its local version too; e.g., Mishra (2016), Karmokar and Roy (2022), Hong and Kim (2023). For results in the context of allocation problems with transfers and quasi-linear utility, see Carroll (2012), Archer and Kleinberg (2014), and Mishra et al. (2016).

Our attempt to relax symmetric manipulation constraints is new in the LGE literature. However, it aligns with a well-explored avenue of research in the broader literature on mechanism design with partially verifiable information. Green and Laffont (1986) assume that the set of admissible messages the agent can transmit to the principal varies with the state of the world and study the relationship between incentive compatibility and implementability; subsequent work focusing on implementation or the revelation principle in a similar vein includes Forges and Koessler (2005), Bull and Watson (2007), Deneckere and Severinov (2008). Milgrom and Roberts (1986) consider a setup where a seller's assertions about product quality are limited to some extent, asymmetrically

[10]Hong and Kim (2023) take a similar approach of adding unanimity but restrict attention to deterministic rules. Thus, their condition of sparsely connected domains without restoration is a sufficient condition for uLGE. They also analyze the relationship between strategy-proofness and local ordinal Bayesian incentive compatibility with respect to generic priors.

across types. In an information revelation game by Okuno-Fujiwara et al. (1990), players can choose among certifiable statements only, which are specified exogenously but not necessarily symmetrically. Certain contexts warrant imposing systematically asymmetric incentive constraints. For example, in the screening models by Celik (2006) and Krähmer and Strausz (2024), an agent can understate, but not overstate, his productivity as his claims should be substantiated by hard evidence. In Che and Gale (2000), a buyer's private information concerns not only his valuation of an item on sale but also his budget constraint, preventing him from imitating types with larger budgets. Sher and Vohra (2015) study a general model where an agent's ability to mimic other types follows a tree (hierarchical) structure.

The rest of the paper is organized as follows. We set up the model in Section 2 and present a characterization of LGE for deterministic rules in Section 3. Our necessity and sufficiency results for RLGE are in Section 4. In Section 5, we conclude by briefly discussing some remaining issues. For a smooth passage, we relegate all proofs to Appendix A.

2 The Model

We study a model where a single agent reports his preferences over alternatives and an alternative is chosen based on those preferences. Given our focus on strategy-proofness (dominant strategy incentive compatibility), the single-agent model is without loss of generality. Let $\boldsymbol{A}$ be a finite set of alternatives. To avoid triviality, assume that $|A| \geq 3$. Denote alternatives by a, b, x, y, and so on. The agent has a strict preference relation $\boldsymbol{P}$ over A (i.e., P is an irreflexive, total, asymmetric, and transitive relation over A). For all $a, b \in A$, $\boldsymbol{a\,P\,b}$ means that the agent prefers a to b. Let $\boldsymbol{\mathcal{P}}$ be the set of all such preference relations. For all disjoint $B, C \subseteq A$, write $\boldsymbol{B\,P\,C}$ if for each $b \in B$ and each $c \in C$, $b\,P\,c$. When B or C is a singleton, with $B = \{b\}$ or $C = \{c\}$, we simply write $b\,P\,C$ or $B\,P\,c$.

A **domain** is a non-empty subset $\boldsymbol{\mathcal{D}} \subseteq \mathcal{P}$, from which the agent's prefer-

ences are drawn. Let us call elements of $\mathcal{D}$ **types** and $\mathcal{P}$ the **universal domain**. We assume that $|\mathcal{D}| \geq 3$.[11] A (deterministic) **rule** is a mapping $\varphi : \mathcal{D} \to A$ that associates with each type $P \in \mathcal{D}$ an alternative $\varphi(P) \in A$. The agent's type is private information and he may gain by misrepresenting it. For the reasons explained in Section 1.1, we assume that the agent faces a constraint when lying. À la Green and Laffont (1986), the misrepresentation constraint can be written as a correspondence $\mu : \mathcal{D} \rightrightarrows \mathcal{D}$, where for each $P \in \mathcal{D}$, $\mu(P) \subseteq \mathcal{D}$ is the set of possible misrepresentations for type P and $P \in \mu(P)$. No restriction is placed on μ. The misrepresentation constraint may be asymmetric across types (for some $P, P' \in \mathcal{D}$, $P' \in \mu(P)$ but $P \notin \mu(P')$; Section 1.2 gives an example of this kind) and "large" lies, according to the Kemeny distance (Kemeny, 1959), may be included. Kumar et al. (2021a) conveniently capture the misrepresentation constraint as a network (graph) defined over the domain, which we follow now.[12]

An **environment** is a pair $\langle \mathcal{D}, \mathcal{E} \rangle$, where $\mathcal{D}$ is a set of vertices (types) and $\mathcal{E} \subseteq \mathcal{D}^2$ is a set of *directed* edges. For all $P, P' \in \mathcal{D}$, a (directed) **edge** $(P, P') \in \mathcal{E}$ from P to P' means that type P can misrepresent himself as type P'. Thus, for each $P \in \mathcal{D}$, $\{P' \in \mathcal{D} : (P, P') \in \mathcal{E}\} = \mu(P)$ is the set of possible types that type P can claim to be. Say that P' is a **unidirectional local to $\boldsymbol{P}$** if $(P, P') \in \mathcal{E}$. In figures, we represent $(P, P') \in \mathcal{E}$ by "$P \to P'$" (i.e., an arrow going from P to P'); and if $(P, P') \in \mathcal{E}$ and $(P', P) \in \mathcal{E}$, we use "$P \longleftrightarrow P'$" and say that P and P' are (bidirectional) **locals**.

A natural way of defining an environment is to declare types to be locals if they are the closest in the sense of the Kemeny distance (Kemeny, 1959). For all $P, P' \in \mathcal{D}$, let $\boldsymbol{P \,\Delta\, P'}$ be the set of unordered pairs $\{a, b\} \subseteq A$ such that P and P' have the opposite preferences over a and b.

Example 1 (*Adjacency environments*)**.** Given $P, P' \in \mathcal{D}$, define the Kemeny dis-

[11] When $|\mathcal{D}| = 2$, our main results change slightly. Theorems 2 and 3 still hold. The equivalence in Theorem 1 no longer holds and we have the following: (i) Property DL implies LGE and the converse fails; and (ii) LGE implies strong connectedness and the converse fails.

[12] Though in a different context, Sher and Vohra (2015) introduce a similar concept, which they call the "incentive graph".

tance between P and P' as $|P \triangle P'|$. Say that P and P' are **Kemeny-adjacent** if $|P \triangle P'| = 1$. An **adjacency environment** is given by $\langle \mathcal{D}, \mathcal{E}^{\text{adj}} \rangle$, where $\mathcal{E}^{\text{adj}} \equiv \{(P, P') \in \mathcal{D}^2 : |P \triangle P'| = 1\}$, so that local types are precisely Kemeny-adjacent ones. Adjacency environments are studied in Carroll (2012), Sato (2013a), Mishra (2016), Cho (2016), Kumar et al. (2021b), and Hong and Kim (2023). $\triangle$

While the Kemeny distance is intuitive and has some axiomatic foundation[13], formulating the notion of locals on its basis reduces applicability of LGE results; e.g., some interesting domains, such as those of separable and multi-dimensional single-peaked preferences, are not covered. Kumar et al. (2021a) address this issue by allowing $\mathcal{E}$ to be more general, to include pairs of types that are not necessarily Kemeny-adjacent.

Example 2 (*Symmetry environments*)**.** A **symmetry environment** is given by $\langle \mathcal{D}, \mathcal{E}^{\text{sym}} \rangle$, where $\mathcal{E}^{\text{sym}}$ satisfies the following **symmetry condition**: for all $P, P' \in \mathcal{D}$, $(P, P') \in \mathcal{E}^{\text{sym}}$ implies $(P', P) \in \mathcal{E}^{\text{sym}}$. For $(P, P') \in \mathcal{E}^{\text{sym}}$, it is possible to have $|P \triangle P'| \geq 2$. Clearly, for an adjacency environment $\langle \mathcal{D}, \mathcal{E}^{\text{adj}} \rangle$, $\mathcal{E}^{\text{adj}}$ satisfies the symmetry condition. Thus, the adjacency environments constitute a proper subfamily of the symmetry environments. We refer to environments violating the symmetry condition as **asymmetry environments**. $\triangle$

Now we define incentive properties. Our notion of incentive compatibility is dominant strategy incentive compatibility, also known as strategy-proofness. Worded in the single-agent context, it requires that the agent should never benefit from reporting any type in the domain.

Strategy-proofness: For all $P, P' \in \mathcal{D}$, $\varphi(P) \, P \, \varphi(P')$ or $\varphi(P) = \varphi(P')$.

As a weakening of strategy-proofness, we may require that a rule be immune to local misrepresentations, where localness is specified by the environment $\langle \mathcal{D}, \mathcal{E} \rangle$.

Local Strategy-proofness: For all $P, P' \in \mathcal{D}$ with $(P, P') \in \mathcal{E}$, $\varphi(P) \, P \, \varphi(P')$ or $\varphi(P) = \varphi(P')$.

[13]Kemeny (1959) characterizes the Kemeny distance and his result is further refined by Can and Storcken (2018).

By definition, a locally strategy-proof rule need not be strategy-proof. However, as Carroll (2012) and Sato (2013a) note, strategy-proofness follows from local strategy-proofness in a broad class of environments. Our objective is to scrutinize the structure of those environments.

Definition 1. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies the **local-global equivalence (LGE)** if each locally strategy-proof rule on $\mathcal{D}$ is strategy-proof.

Existing studies show that a simple necessary condition for LGE concerns the ability to travel from a type to another in $\mathcal{D}$ using only the edges in $\mathcal{E}$ (Sato, 2013a; Kumar et al., 2021a). Given $P, P' \in \mathcal{D}$, a **path from P to P'** is a sequence $\boldsymbol{\pi \equiv (P_1, \cdots, P_k)}$ of distinct vertices in $\mathcal{D}$ such that $P_1 = P$, $P_k = P'$, and for each $j \in \{1, \cdots, k-1\}$, $(P_j, P_{j+1}) \in \mathcal{E}$. Denote by $\boldsymbol{\Pi(P, P')}$ the set of all paths from P to P'. Given a path $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$ and distinct alternatives $a, b \in A$, the path is **with $\{a, b\}$-restoration** if there are $h, i, j \in \{1, \cdots, k\}$ with $h < i < j$ such that either (i) $a \, P_h \, b$, $b \, P_i \, a$, and $a \, P_j \, b$; or (ii) $b \, P_h \, a$, $a \, P_i \, b$, and $b \, P_j \, a$ (Sato, 2013a). We say that the path is **without $\{a, b\}$-restoration** if there is no such (h, i, j). A path has a **restoration** if for some $a, b \in A$, it is with $\{a, b\}$-restoration. Also, for $P, P' \in \mathcal{D}$ and $a, b \in A$, say that a **overtakes** b as we go from P to P' if $b \, P \, a$ and $a \, P' \, b$. If a path is without $\{a, b\}$-restoration, overtaking between a and b takes place at most once along the path.

Below we use the following definitions and notation. Let $P, P', P'' \in \mathcal{D}$. Given $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$ and $\pi' \equiv (P'_1, \cdots, P'_h) \in \Pi(P', P'')$, where no type in π and π' is repeated except for P', we write $\boldsymbol{\pi \cup \pi'}$ for the path $(P_1, \cdots, P_k, P'_2, \cdots, P'_h)$ from P to P'' obtained by concatenating π and π'. Also, for $P_i, P_j \in \pi$ with $1 \leq i < j \leq k$, denote by $\boldsymbol{\pi|_{[P_i, P_j]}}$ the subpath $(P_i, \cdots, P_j)$ from P_i to P_j. For all $P, P' \in \mathcal{D}$, P' is the **inverse of P** if for all $a, b \in A$, $a \, P' \, b$ if and only if $b \, P \, a$.

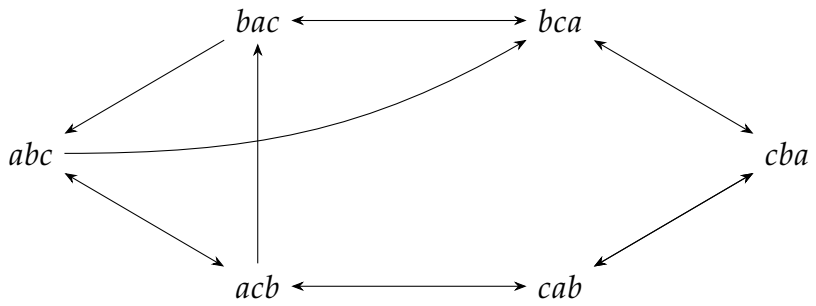

Figure 2: An (asymmetry) environment $\langle \mathcal{D}^*, \mathcal{E}^* \rangle$ satisfying LGE

3 LGE for Deterministic Rules: Characterization

To motivate our conditions characterizing LGE, we first attempt to derive a necessary condition, with the aid of a simple example. Suppose that $A = \{a, b, c\}$, $\mathcal{D}^* = \{abc, acb, bac, bca, cab, cba\}$ is the universal domain[14], and $\mathcal{E}^*$ is given by the edges in Figure 2. Clearly, $\langle \mathcal{D}^*, \mathcal{E}^* \rangle$ violates the symmetry condition because, e.g., $(abc, bac) \notin \mathcal{E}^*$.

One can show that environment $\langle \mathcal{D}^*, \mathcal{E}^* \rangle$ satisfies LGE.[15] Further, it is a "tight" example of LGE in the sense that removing *any* edge in $\mathcal{E}^*$ leads to a failure of LGE. For instance, remove an edge $(abc, bca) \in \mathcal{E}^*$ and consider the new environment $\langle \mathcal{D}^*, \mathcal{E}^* \backslash \{(abc, bca)\} \rangle$ (see panel (a) of Figure 3). Define a rule $\varphi : \mathcal{D}^* \to A$ as $\varphi(abc) = \varphi(acb) = \varphi(cab) = \varphi(cba) = c$ and $\varphi(bac) = \varphi(bca) = b$. Except for type abc, φ is specified in such a way that for each $P \in \mathcal{D}^* \backslash \{abc\}$, $\varphi(P)$ is the alternative that P prefers among b and c. This means that abc is the only type with an incentive to manipulate φ. Local misrepresentations do not give b to type abc since there is no edge from abc to a type to which φ assigns b (namely, bac and bca). Thus, φ is locally strategy-proof. However, φ is not strategy-proof because type abc benefits from reporting bca.

Figure 3 shows seven environments that are obtained by modifying $\langle \mathcal{D}^*, \mathcal{E}^* \rangle$.[16] Each environment misses an edge in $\mathcal{E}^*$, which is indicated by a **green** dashed

[14]We denote by "abc" the preference relation that ranks a first and b second.

[15]This can be proved directly or by applying Theorem 1 below.

[16]There are several more environments one can get by removing an edge in $\mathcal{E}^*$, from a type to another with the same top alternative (e.g., removing $(abc, acb) \in \mathcal{E}^*$). We do not include them here as showing a failure of LGE is more straightforward for them and key implications of LGE emerge from the environments in Figure 3.

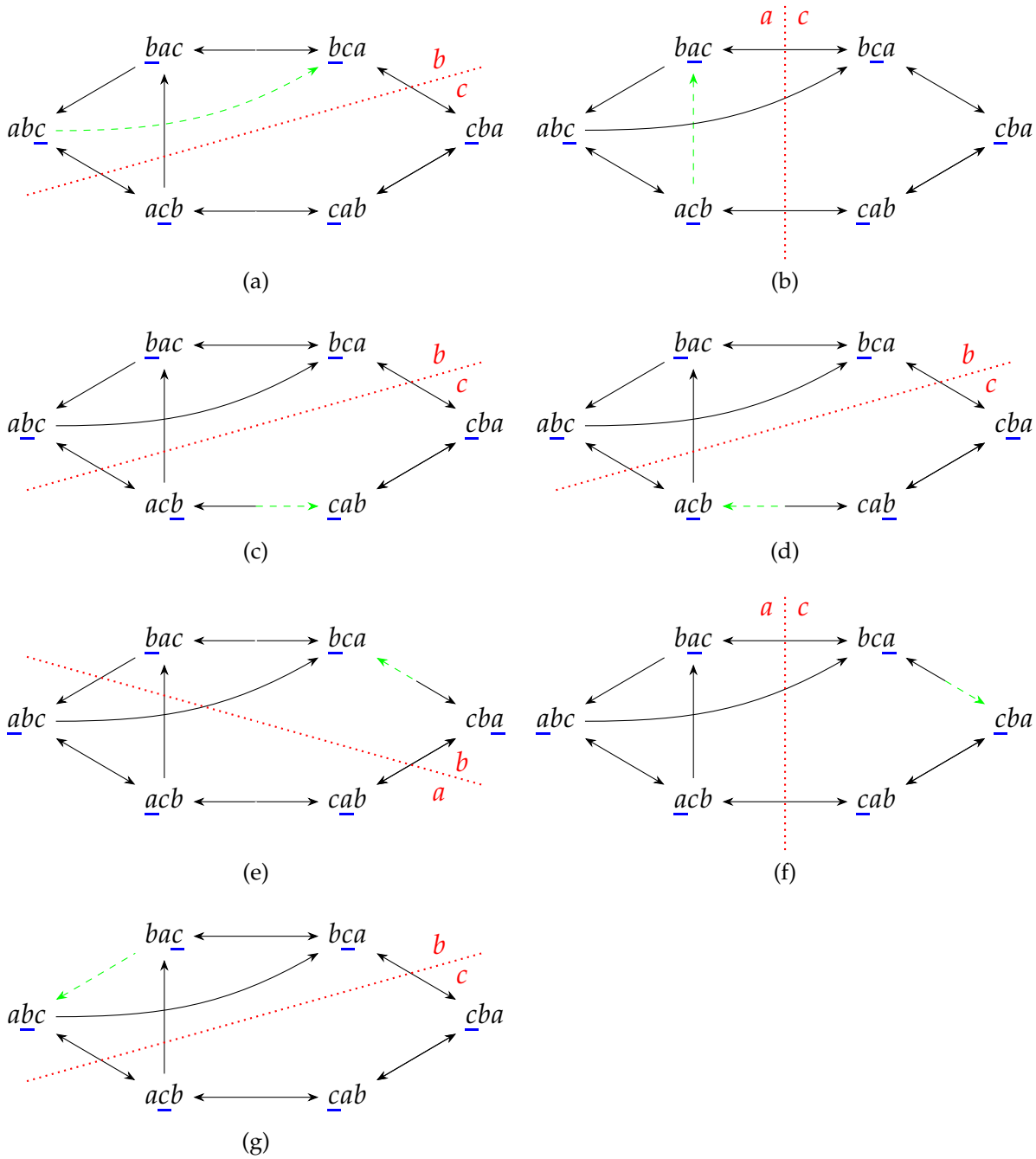

Figure 3: Modifications of environment $\langle \mathcal{D}^*, \mathcal{E}^* \rangle$ in Figure 2 failing LGE

arrow. Each violates LGE and we include in the figure a rule establishing the latter, by underlining in **blue** the alternative the rule chooses for each type. For instance, in panel (b), the rule assigns a to type bac and c to all the other types.

Then what distinguishes environment $\langle \mathcal{D}^*, \mathcal{E}^* \rangle$ from the seven environments in Figure 3? To answer, let us begin with a condition that has proved fundamental for LGE: connectedness. Say that an environment $\langle \mathcal{D}, \mathcal{E} \rangle$ is **connected** if for all $P, P' \in \mathcal{D}$, there is a path from P to P', i.e., $\Pi(P, P') \neq \emptyset$. For example, environment $\langle \mathcal{D}^*, \mathcal{E}^* \rangle$ is connected. Sato (2013a) and Kumar et al. (2021a) show that in adjacency and symmetry environments, connectedness is necessary for LGE. The latter necessity continues to be valid for general environments. However, similarly to what has been found by Sato (2013a) and Kumar et al. (2021a), connectedness is far from sufficient, as confirmed by the environments in Figure 3, and therefore cannot be the defining feature of environment $\langle \mathcal{D}^*, \mathcal{E}^* \rangle$.

Let us take the environment in panel (a) of Figure 3 and partition the domain into two subsets, according to which among alternatives b and c is preferred. The types above the **red** dotted line prefer b and those below it prefer c. The lower "half-space", consisting of acb, cab, and cba, is connected in its own right: we can always traverse from a type to another in the lower half-space, without leaving it. On the other hand, the upper half-space, consisting of abc, bac, and bca, is not connected (since there is no path from abc to bca within the half-space) and our example rule showing that this environment violates LGE exploits such disconnectedness. Similar observations hold for the other six environments in Figure 3. We may choose an appropriate pair of alternatives so that one of the resulting half-spaces is not connected. By contrast, all half-spaces of environment $\langle \mathcal{D}^*, \mathcal{E}^* \rangle$ are connected, regardless of how we partition it.

Indeed, connectedness of "subenvironments" characterizes the boundary between LGE and its failure. A **subdomain** is a non-empty subset of domain $\mathcal{D}$. Given a subdomain $\mathcal{D}' \subseteq \mathcal{D}$, let $\boldsymbol{\mathcal{E}}|_{\mathcal{D}'} \equiv \{(P, P') \in \mathcal{E} : P, P' \in \mathcal{D}'\}$ be the restriction of $\mathcal{E}$ to the vertices in $\mathcal{D}'$. In relation to $\langle \mathcal{D}, \mathcal{E} \rangle$, we refer to $\langle \mathcal{D}', \mathcal{E}|_{\mathcal{D}'} \rangle$ as a **subenvironment**. For all $P, P' \in \mathcal{D}'$, a path $\pi \equiv (P_1, \cdots, P_k)$ from P to P' is **in**

$\langle \boldsymbol{\mathcal{D}'}, \boldsymbol{\mathcal{E}}|_{\boldsymbol{\mathcal{D}'}} \rangle$ if $P_1, \cdots, P_k \in \mathcal{D}'$ and for each $j \in \{1, \cdots, k-1\}$, $(P_j, P_{j+1}) \in \mathcal{E}|_{\mathcal{D}'}$. Denote by $\boldsymbol{\Pi(P, P', \mathcal{E}|_{\mathcal{D}'})}$ the set of all paths from P to P' in $\langle \mathcal{D}', \mathcal{E}|_{\mathcal{D}'} \rangle$. Say that $\langle \mathcal{D}', \mathcal{E}|_{\mathcal{D}'} \rangle$ is **connected** if for all $P, P' \in \mathcal{D}'$, there is a path from P to P' in $\langle \mathcal{D}', \mathcal{E}|_{\mathcal{D}'} \rangle$, i.e., $\Pi(P, P', \mathcal{E}|_{\mathcal{D}'}) \neq \emptyset$.[17] Also, for each $B \subseteq A$ and each $a \in B$, define $\boldsymbol{\mathcal{D}(a, B)} \equiv \{P \in \mathcal{D} : P$ most prefers a in $B\}$. To use this notation, the half-spaces we mentioned above for panel (a) in Figure 3 are $\mathcal{D}(b, \{b, c\})$ and $\mathcal{D}(c, \{b, c\})$. If $B = \{a_1, \cdots, a_k\}$, then the collection $\{\mathcal{D}(a_j, B)\}_{j=1,\cdots,k}$ partitions $\mathcal{D}$ (some sets in the collection may be empty). Now we are ready to define our key condition.

Definition 2. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ is **strongly connected** if for each $B \subseteq A$ with $|B| \geq 2$ and each $a \in B$, the subenvironment $\langle \mathcal{D}(a, B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$ is connected.[18,19]

The following example illustrates strong connectedness.

Example 3 (*Illustrating strong connectedness*)**.** Let us illustrate implications of strong connectedness using three alternatives, referring to Figure 4. Let $a, b, c \in A$ be distinct. The pair $\{a, b\}$ yields two half-spaces, $\langle \mathcal{D}(a, \{a, b\}), \mathcal{E}|_{\mathcal{D}(a,\{a,b\})} \rangle$ and $\langle \mathcal{D}(b, \{a, b\}), \mathcal{E}|_{\mathcal{D}(b,\{a,b\})} \rangle$. The border between the two half-spaces, which we call the ***a*-*b* border**, is indicated by a **red** dotted line in the figure. Similarly, the pair $\{a, c\}$ leads to half-spaces $\langle \mathcal{D}(a, \{a, c\}), \mathcal{E}|_{\mathcal{D}(a,\{a,c\})} \rangle$ and $\langle \mathcal{D}(c, \{a, c\}), \mathcal{E}|_{\mathcal{D}(c,\{a,c\})} \rangle$ and the *a*-*c* border in a **blue** dotted line, and the pair $\{b, c\}$ to half-spaces $\langle \mathcal{D}(b, \{b, c\}), \mathcal{E}|_{\mathcal{D}(b,\{b,c\})} \rangle$ and $\langle \mathcal{D}(c, \{b, c\}), \mathcal{E}|_{\mathcal{D}(c,\{b,c\})} \rangle$ and the *b*-*c* border in a **green** dotted line. Strong connectedness says that in each of these six half-spaces, it is possible to go from a type to another in that half-space, without crossing the relevant border.

[17] With an abuse of language, we also take $\langle \mathcal{D}', \mathcal{E}|_{\mathcal{D}'} \rangle$ to be connected if $|\mathcal{D}'| = 1$ or 0.

[18] It is known that on some domains, every strategy-proof and "unanimous" rule satisfies the "tops-only property". Chatterji and Zeng (2018) provide sufficient conditions on domains under which this is true. While Chatterji and Zeng (2018) have a different focus than ours, a condition of theirs, called the interior property, bears a resemblance to strong connectedness (and is weaker). It turns out that if an adjacency environment $\langle \mathcal{D}, \mathcal{E}^{\text{adj}} \rangle$ satisfies LGE, then $\mathcal{D}$ is a tops-only domain.

[19] One may define a weaker version of strong connectedness by confining its requirement to sets $B \subseteq A$ with $|B| = 2$. The resulting property is investigated in Section 3.1.

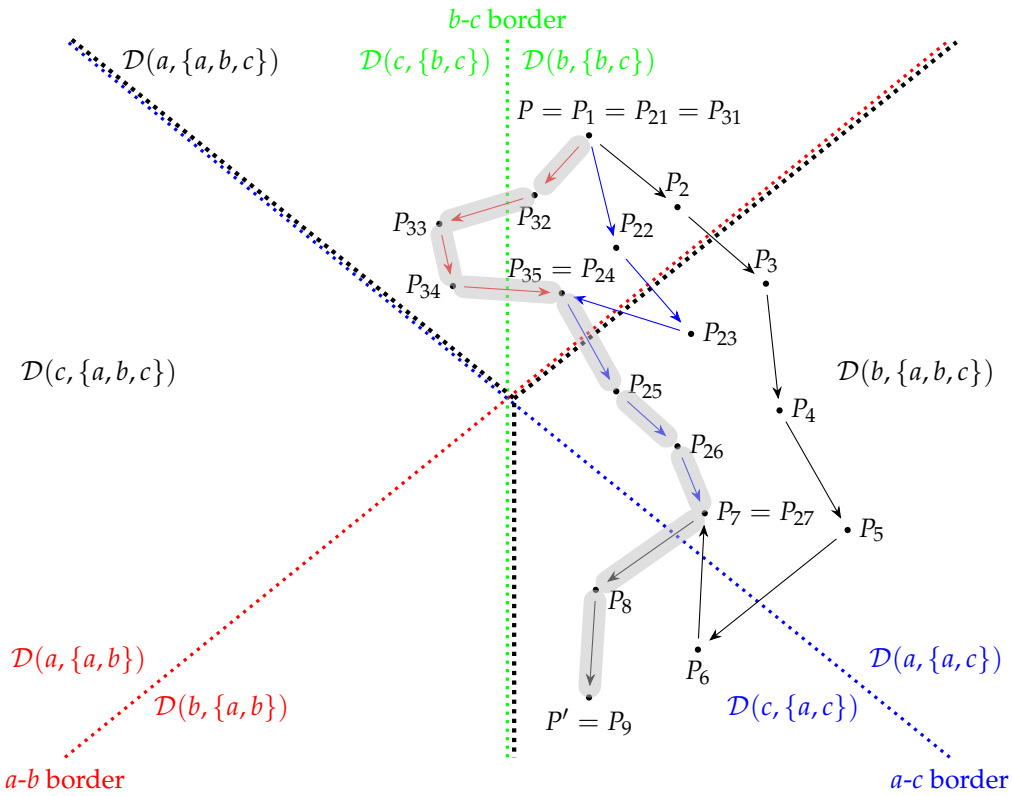

Figure 4: Illustrating implications of strong connectedness using three alternatives (Examples 3 and 4).

Now let us consider a triple $\{a,b,c\}$. Associated with the triple are "third-spaces" $\langle \mathcal{D}(a,\{a,b,c\}), \mathcal{E}|_{\mathcal{D}(a,\{a,b,c\})}\rangle$, $\langle \mathcal{D}(b,\{a,b,c\}), \mathcal{E}|_{\mathcal{D}(b,\{a,b,c\})}\rangle$, and $\langle \mathcal{D}(c,\{a,b,c\}), \mathcal{E}|_{\mathcal{D}(c,\{a,b,c\})}\rangle$. The border dividing them are represented by thick black dotted lines. By strong connectedness again, we can freely traverse between types within a third-space. $\triangle$

Several well-known environments satisfy not only strong connectedness but also an even stronger property we introduce in Section 4 below. See Example 10 therein for details.

Significant implications of strong connectedness emerge when one attempts to go beyond a given subenvironment. If we start from a type in $\langle \mathcal{D}(a,\{a,b\}), \mathcal{E}|_{\mathcal{D}(a,\{a,b\})}\rangle$, say, does there always exist a path that crosses the a-b border and reaches other types in $\langle \mathcal{D}(b,\{a,b\}), \mathcal{E}|_{\mathcal{D}(b,\{a,b\})}\rangle$? If so, how many border crossings are needed? The first question pertains to connectedness of the environment under consideration. Our result below answers it in the affirmative.

Proposition 1. *If an environment $\langle \mathcal{D}, \mathcal{E} \rangle$ is strongly connected, then it is connected.*

Now we turn to the second question, pertaining to the number of border crossings required. The following result says that for types with different top alternatives, say a and b, in B, regardless of whether $B = \{a, b\}$ or $B \supsetneq \{a, b\}$, we can traverse between them with only one crossing of the a-b border. If translated to the no-restoration language, it means that there is no $\{a, b\}$-restoration.

Proposition 2 ("Single-crossing of borders")**.** *Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be a strongly connected environment.*

(1) *Let $a, b \in A$ be distinct. Then for each $P \in \mathcal{D}(a, \{a, b\})$ and each $P' \in \mathcal{D}(b, \{a, b\})$, there is a path $\pi \in \Pi(P, P')$ without $\{a, b\}$-restoration.*

(2) *Let $B \subseteq A$ be non-empty and $a, b \in B$ be distinct. Then for each $P \in \mathcal{D}(a, B)$ and each $P' \in \mathcal{D}(b, B)$, there is a path $\pi \in \Pi(P, P')$ without $\{a, b\}$-restoration.*

While strong connectedness has to do with connectedness of subenvironments, Proposition 2 hints at its potential relationship with other properties of environments that are based on the notion of restoration. The latter properties include (i) the one identified by Sato (2013a) for adjacency environments, and (ii) the one identified by Kumar et al. (2021a) for symmetry environments. Now we define them.

Sato (2013a) is the first to note the relevance of restoration on LGE. His condition, termed the no-restoration property, says that for all types $P, P' \in \mathcal{D}$, there should be a path that has no restoration for any pair of alternatives.

Definition 3. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies the **no-restoration property** if for all $P, P' \in \mathcal{D}$, there is $\pi \in \Pi(P, P')$ such that for all $a, b \in A$, π is without $\{a, b\}$-restoration.[20]

There is another condition that substantially weakens the no-restoration property and is a complete characterization of the boundary between LGE and its failure. To define it, for each $P \in \mathcal{D}$ and each $a \in A$, let $\boldsymbol{L(P, a)} \equiv \{b \in$

[20]Sato (2013a) also provides a sufficient condition slightly weaker than the no-restoration property.

$A : a\,P\,b\}$ be the (strict) **lower contour set of P at a**. Kumar et al. (2021a) introduce the **"lower-contour-set no-restoration property"**, or simply **"Property L"**, which requires that for all types P and P' and each alternative a, there exists a path between P and P' such that for each $b \in L(P, a)$, the path is without $\{a, b\}$-restoration. In symmetry environments, the required path allows one to go from P to P' and also from P' to P. However, to accommodate general environments with asymmetric manipulation, we should indicate the direction of the path too and Property L can be adapted as follows.

Definition 4. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies the **directed lower-contour-set no-restoration property**, or simply **Property DL**, if for all $P, P' \in \mathcal{D}$ and each $a \in A$, there is a path $\pi \in \Pi(P, P')$ such that for each $b \in L(P, a)$, π is without $\{a, b\}$-restoration.

Property L by Kumar et al. (2021a) obtains if we restrict the above definition to symmetry environments $\langle \mathcal{D}, \mathcal{E}^{\text{sym}} \rangle$. While Property DL is a natural analog of Property L in our setup, it has the potential to be quite weak relative to Property L. Suppose that $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$ is the path whose existence is required by Property DL. Local strategy-proofness implies that for each $j \in \{1, \cdots, k-1\}$, P_j should not gain by reporting P_{j+1}. If the environment were a symmetry one, we may exploit the fact that P_{j+1} is not better off reporting P_j.[21] However, in asymmetry environments, $(P_{j+1}, P_j) \in \mathcal{E}$ is not guaranteed and when it is not the case, we should turn to a new path from P_{j+1} to P_j, which may be unwieldy. Consequently, local strategy-proofness implies a more limited set of constraints in asymmetry environments and LGE is more likely to fail.

However, as we show below, LGE, strong connectedness, and Property DL are equivalent. While a proof of the equivalence is complicated, we may gain intuition for the relationship among them by examining a three-alternative example.

[21] This type of argument is common to several proofs on LGE; e.g., Carroll (2012), Sato (2013), and Kumar et al. (2021a).

Example 4 (*Illustrating that strong connectedness implies Property DL*)**.** Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be a strongly connected environment. To show that $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies Property DL, we need to find, for all $P, P' \in \mathcal{D}$ and each $a \in A$, a path from P to P' with some special property. For the sake of illustration, suppose that for the chosen P, P', and a, we have $L(P, a) = \{b, c\}$, $a\, P\, b\, P\, c$, and $b\, P'\, c\, P'\, a$. With the focus on $\{a, b, c\}$, we may plot P and P' in Figure 4. Note that $P \in \mathcal{D}(a, \{a, b, c\}) \cap \mathcal{D}(b, \{b, c\})$ and $P' \in \mathcal{D}(b, \{a, b, c\}) \cap \mathcal{D}(c, \{a, c\})$. Property DL requires the existence of a path from P to P' without $\{a, b\}$- and $\{a, c\}$-restoration. Given $a\, P\, b$ and $b\, P'\, a$, this means that the path should cross the a-b border exactly once. Similarly, the path should cross the a-c border exactly once. We now construct such a path appealing to strong connectedness.

First, by Proposition 1, $\langle \mathcal{D}, \mathcal{E} \rangle$ is connected, so that there is a path, say $(P_1, P_2, \cdots, P_9)$ (in black), from P to P'. If it crosses each of the a-b and a-c borders once, we are done. Otherwise, search for the last crossing of the a-b and a-c borders in path $(P_1, P_2, \cdots, P_9)$. Suppose that as in the figure, the last crossing is over the a-c border, as we go from P_7 to P_8 (i.e., P_7 is defined to be the last type in path $(P_1, P_2, \cdots, P_9)$ that prefers a to c). Then we keep only the subpath (P_7, P_8, P_9) and find other ways of reaching P_7 from P.

Since $P, P_7 \in \mathcal{D}(a, \{a, c\})$ and the subenvironment $\langle \mathcal{D}(a, \{a, c\}), \mathcal{E}|_{\mathcal{D}(a,\{a,c\})} \rangle$ is connected, there is a path, say $(P_{21}, P_{22}, \cdots, P_{27})$ (in **blue**), from P to P_7 in $\langle \mathcal{D}(a, \{a, c\}), \mathcal{E}|_{\mathcal{D}(a,\{a,c\})} \rangle$. This path does not cross the a-c border but may cross the a-b border more than once, as is the case in the figure. Then we look for the last crossing of the a-b border in path $(P_{21}, P_{22}, \cdots, P_{27})$. Assume that as in the figure, the latter occurs when we go from P_{24} to P_{25} (i.e., P_{24} is defined to be the last type in path $(P_{21}, P_{22}, \cdots, P_{27})$ that prefers a to b). If we concatenate paths $(P_{24}, P_{25}, P_{26}, P_{27})$ and (P_7, P_8, P_9), each of the a-b and a-c borders is crossed exactly once.

Now it remains to link P to P_{24} without crossing the a-b and a-c borders. Since $(P_{21}, P_{22}, \cdots, P_{27})$ is a path in $\langle \mathcal{D}(a, \{a, c\}), \mathcal{E}|_{\mathcal{D}(a,\{a,c\})} \rangle$, $P_{24} \in \mathcal{D}(a, \{a, b, c\})$. Together with $P \in \mathcal{D}(a, \{a, b, c\})$, connectedness of $\langle \mathcal{D}(a, \{a, b, c\}), \mathcal{E}|_{\mathcal{D}(a,\{a,b,c\})} \rangle$

implies that there is a path, say $(P_{31}, P_{32}, \cdots, P_{35})$ (in **red**) from P to P_{24} in $\langle \mathcal{D}(a, \{a, b, c\}), \mathcal{E}|_{\mathcal{D}(a,\{a,b,c\})} \rangle$. The latter path never crosses the *a*-*b* and *a*-*c* borders (although it may cross the *b*-*c* border, possibly multiple times). Then the concatenated path $(P_{31}, P_{32}, \cdots, P_{35}) \cup (P_{24}, P_{25}, P_{26}, P_{27}) \cup (P_7, P_8, P_9)$ (highlighted in **gray**) is the desired path. Our proof that strong connectedness implies Property DL generalizes this intuition to cover all possibilities. △

Now we state our first main result. It says that LGE, strong connectedness, and Property DL are equivalent.

Theorem 1. *For each environment $\langle \mathcal{D}, \mathcal{E} \rangle$, the following statements are equivalent.*

(1) *$\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies LGE.*

(2) *$\langle \mathcal{D}, \mathcal{E} \rangle$ is strongly connected.*

(3) *$\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies Property DL.*

Remark 1. The theorem above allows two interpretations that are new in the LGE literature. One is that LGE is characterized by strong connectedness. As we saw in Figures 3 and 4, strong connectedness has a simple visual representation. Since LGE is characterized by strong connectedness, it can serve as a tractable test for LGE. The second interpretation comes from the equivalence of strong connectedness and Property DL. With the exception of connectedness, existing sufficient and necessary conditions for LGE analyze environments from the no-restoration perspective.[22] The no-restoration property by Sato (2013a) requires that there be a path without restoration for any pair of alternatives; and Property L by Kumar et al. (2021a) requires that there be a path without restoration for a given alternative and its lower contour set. By contrast, strong connectedness revolves around the connectedness idea. An environment is decomposed into various subenvironments, each of which consists of types sharing a feature in preferences and is itself connected within. The equivalence of strong connectedness and Property DL indicates that the no-restoration and connectedness perspectives, if refined enough, have a common

[22]This is the case for Property L, the no-restoration property, and the pairwise no-restoration property (defined in Section 3.1).

root and are just two different ways of approaching LGE. $\triangle$

Restricting Theorem 1 to symmetry environments, we obtain the following corollary.

Corollary 1 (Kumar et al., 2021a)**.** *Let $\langle \mathcal{D}, \mathcal{E}^{sym} \rangle$ be a symmetry environment. Then $\langle \mathcal{D}, \mathcal{E}^{sym} \rangle$ satisfies LGE if and only if it satisfies Property L.*

So far, we have introduced several properties of environments and investigated their implications. Figure 1 in Section 1.3 summarizes the logical relationship among them.

3.1 A Simpler Condition for LGE

Strong connectedness requires each subenvironment $\langle \mathcal{D}(a, B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$ to be connected. One may define a weaker version of strong connectedness by confining its requirement to sets $B \subseteq A$ with $|B| = 2$. In this subsection, we show that the weakening, in general, is intermediate in strength between strong connectedness and connectedness; and that for some class of environments, it is equivalent to strong connectedness, and hence to LGE. Therefore, the weakening can serve as a simpler test for LGE.

First, let us introduce a weakening of strong connectedness.

Definition 5. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ is **pairwise connected** if for all distinct $a, b \in A$, the subenvironment $\langle \mathcal{D}(a, \{a, b\}), \mathcal{E}|_{\mathcal{D}(a,\{a,b\})} \rangle$ is connected.

Pairwise connectedness involves verifying connectedness of all half-spaces. As there are "many" half-spaces, it is reasonable to ask if this requirement is sufficient for strong connectedness. Our example below answers the question in the negative.

Example 5 (*Pairwise connectedness does not imply strong connectedness*)**.** Assume that $A = \{a, b, c\}$ and consider an (asymmetry) environment $\langle \mathcal{D}, \mathcal{E} \rangle$ in Figure 5, where $\mathcal{D}$ is the universal domain and $\mathcal{E}$ is given by the edges in the figure. It is simple to check that for all distinct $x, y \in A$, the subenvironment

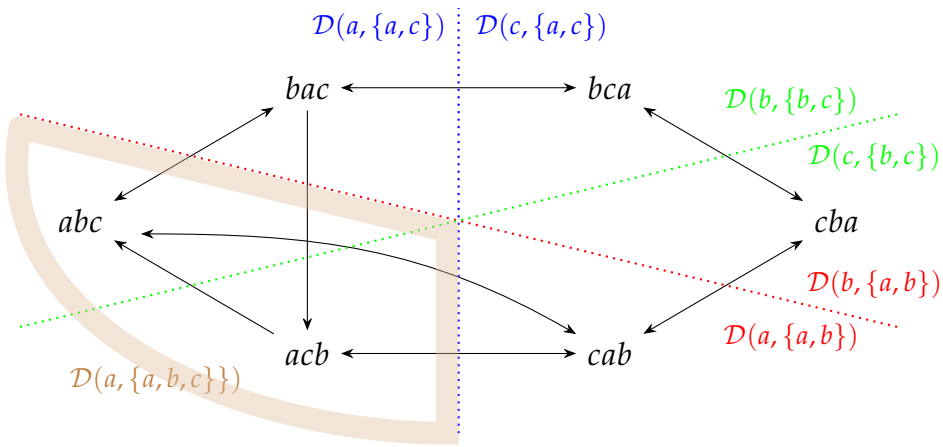

Figure 5: An environment satisfying pairwise connectedness but not strong connectedness

$\langle \mathcal{D}(x,\{x,y\}), \mathcal{E}|_{\mathcal{D}(x,\{x,y\})} \rangle$ is connected, so that $\langle \mathcal{D}, \mathcal{E} \rangle$ is pairwise connected. However, $\langle \mathcal{D}, \mathcal{E} \rangle$ is not strongly connected because $\langle \mathcal{D}(a,A), \mathcal{E}|_{\mathcal{D}(a,A)} \rangle$ is not connected. $\triangle$

Since LGE is characterized by strong connectedness, Example 5 implies that pairwise connectedness is only necessary for LGE. Then how does it relate to other necessary conditions, say connectedness? Recall that by Proposition 1, strong connectedness implies connectedness. In fact, the proof of the proposition exploits only the fact that subenvironments of the kind $\langle \mathcal{D}(a,\{a,b\}), \mathcal{E}|_{\mathcal{D}(a,\{a,b\})} \rangle$ are connected. This is precisely the requirement of pairwise connectedness. Thus, pairwise connectedness implies connectedness.

Pairwise connectedness, however, turns out equivalent to strong connectedness when a domain satisfies some condition. To define the latter, we draw on the notion of value restriction, introduced by Sen (1966) for preference aggregation. Let $P \in \mathcal{D}$. Given a triple of alternatives, an alternative is assigned the **best value** by P if P most prefers it among the triple; it is assigned the **worst value** by P if P least prefers it among the triple. A domain $\mathcal{D}$ satisfies **best-value restriction** if for any triple of alternatives, there is an alternative that is never assigned the best value by any type in $\mathcal{D}$; i.e., for each triple $a, b, c \in A$ of distinct alternatives, there is an alternative, say a, such that for each $P \in \mathcal{D}$, $b\,P\,a$ or $c\,P\,a$. Similarly, $\mathcal{D}$ satisfies **worst-value restriction** if for any triple of

alternatives, there is an alternative that is never assigned the worst value by any type in $\mathcal{D}$; i.e., for each triple $a, b, c \in A$ of distinct alternatives, there is an alternative, say a, such that for each $P \in \mathcal{D}$, $a \; P \; b$ or $a \; P \; c$. Below are examples of domains satisfying best- and worst-value restrictions.

Example 6 (*Domains of single-peaked and single-dipped types*)**.** Let $\succ$ be a linear order on A (with the associated weak order denoted $\succsim$). A type $P \in \mathcal{D}$ is **single-peaked** (with respect to $\succ$) if there is a unique alternative, $p(P) \in A$, which we call the "peak alternative for P", such that for all $a, b \in A$ with $a \prec b \precsim p(P)$ or $p(P) \precsim b \prec a$, $b \; P \; a$ (Black, 1958). Let $\boldsymbol{\mathcal{D}_{sp}}$ be the domain of all single-peaked types. On the other hand, $P \in \mathcal{D}$ is **single-dipped** (with respect to $\succ$) if there is a unique alternative, $d(P) \in A$, which we call the "dip alternative for P", such that for all $a, b \in A$ with $a \prec b \precsim d(P)$ or $d(P) \precsim b \prec a$, $a \; P \; b$ (Vickrey, 1960). Let $\boldsymbol{\mathcal{D}_{sd}}$ be the domain of all single-dipped types. Each domain $\mathcal{D} \subseteq \mathcal{D}_{sp}$ satisfies worst-value restriction while each domain $\mathcal{D} \subseteq \mathcal{D}_{sd}$ satisfies best-value restriction (Inada, 1964).[23] $\triangle$

Now we show that for environments with the domain satisfying best-value restriction, pairwise connectedness is equivalent to strong connectedness.

Proposition 3. *Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment with $\mathcal{D}$ satisfying best-value restriction. Then $\langle \mathcal{D}, \mathcal{E} \rangle$ is pairwise connected if and only if it is strongly connected.*

However, worst-value restriction, or even the more demanding requirement of single-peakedness, does not guarantee a similar equivalence.

Example 7 (*Under the assumption of worst-value restriction, pairwise connectedness does not imply strong connectedness*)**.** Let $A = \{a, b, c\}$ and assume that the linear order $\succ$ on A satisfies $c \prec b \prec a$. Then the single-peaked domain is $\mathcal{D}_{sp} = \{abc, bac, bca, cba\}$. Consider an environment $\langle \mathcal{D}_{sp}, \mathcal{E} \rangle$ in Figure 6. One can easily verify that $\langle \mathcal{D}_{sp}, \mathcal{E} \rangle$ is pairwise connected. However, it is not strongly connected because $\langle \mathcal{D}(b, A), \mathcal{E}|_{\mathcal{D}(b,A)} \rangle$ is not connected. $\triangle$

[23]The notions of single-peakedness and single-dippedness are more restrictive than worst- and best-value restriction, respectively.

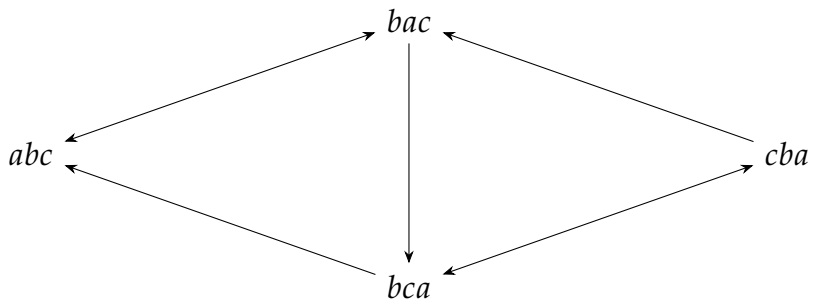

Figure 6: An environment showing that under worst-value restriction, pairwise connectedness does not imply strong connectedness (Example 7).

Pairwise connectedness has a counterpart among properties based on the notion of no-restoration. Sato (2013a) identifies a necessary condition for LGE that is stronger than connectedness. It is a weaker version of the no-restoration property, applying only to a fixed pair of alternatives.

Definition 6. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies the **pairwise no-restoration property**[24] if for all $P, P' \in \mathcal{D}$ and all $a, b \in A$, there is a path $\pi \in \Pi(P, P')$ without $\{a, b\}$-restoration.

Pairwise connectedness turns out equivalent to the pairwise no-restoration property.

Proposition 4. *An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ is pairwise connected if and only if it satisfies the pairwise no-restoration property.*

This proposition parallels Theorem 1 in the following sense. Theorem 1 shows the equivalence of strong connectedness and Property DL, which respectively views environments from the connectedness and no-restoration standpoints (see Remark 1). The proposition above also shows how a connectedness notion, pairwise connectedness, can be translated into a no-restoration notion, the pairwise no-restoration property.

Combining Propositions 3 and 4 and Theorem 1, we obtain the following corollary.

[24]This property was introduced by Sato (2013a) but was not named there. The term "pairwise no-restoration property" is due to Kumar et al. (2021a).

Corollary 2. *For each environment $\langle \mathcal{D}, \mathcal{E} \rangle$ with $\mathcal{D}$ satisfying best-value restriction, the following properties are equivalent: LGE, strong connectedness, pairwise connectedness, Property DL, and the pairwise no-restoration property.*

4 LGE for Random Rules: Necessity and Sufficiency

Now we turn to the issue of LGE for "random" rules (which can choose a lottery over alternatives). In addition to the concepts and notation introduced in previous sections, we use the following. Let $P \in \mathcal{D}$. For each $a \in A$, let $\boldsymbol{U(P,a)} \equiv \{b \in A : b\, P\, a\}$ be the (strict) **upper contour set of $\boldsymbol{P}$ at $\boldsymbol{a}$**. Also, for each $B \subseteq A$, B is an **upper contour set of $\boldsymbol{P}$** if for some $a \in A$, $U(P,a) = B$. For each $k \in \{1, \cdots, |A|\}$, let $\boldsymbol{r_k(P)}$ be the k-th ranked alternative according to P. For each $B \subseteq A$, let $\boldsymbol{P|_B}$ be the restriction of P to B; and for each $k \in \{1, \cdots, |B|\}$, let $\boldsymbol{r_k(P|_B)}$ be the alternative that P ranks k-th within B.

A **lottery** is a probability distribution $\boldsymbol{\lambda} \equiv (\lambda_a)_{a \in A}$ over alternatives, where for each $a \in A$, $\lambda_a \in [0,1]$ is the probability of alternative a being chosen and $\sum_{a \in A} \lambda_a = 1$. Let $\boldsymbol{\Delta(A)}$ be the set of all lotteries over A. For each $\lambda \in \Delta(A)$, let $\mathbf{supp}(\boldsymbol{\lambda}) \equiv \{a \in A : \lambda_a > 0\}$ be the **support of lottery $\boldsymbol{\lambda}$**. We sometimes use the following simpler representation of lotteries: e.g., for $a, b, c \in A$, $\lambda = \frac{1}{2}a + \frac{1}{3}b + \frac{1}{6}c$ means the lottery $\lambda \in \Delta(A)$ such that $\lambda_a = \frac{1}{2}$, $\lambda_b = \frac{1}{3}$, $\lambda_c = \frac{1}{6}$, and for each $d \in A \backslash \{a, b, c\}$, $\lambda_d = 0$. A **random rule** is a mapping $\varphi : \mathcal{D} \to \Delta(A)$, associating with each type $P \in \mathcal{D}$ a lottery $\varphi(P) \equiv (\varphi_a(P))_{a \in A} \in \Delta(A)$. For each $P \in \mathcal{D}$ and each $B \subseteq A$, define $\boldsymbol{\varphi_B(P)} \equiv \sum_{b \in B} \varphi_b(P)$.

When a random rule is in place, the agent equipped with (ordinal) preferences over alternatives need to evaluate lotteries defined over those alternatives. Following a prevalent approach in the ordinal mechanism design literature (Gibbard, 1977; Bogomolnaia and Moulin, 2001), we assume that the agent compares lotteries using first-order stochastic dominance. For each $P \in \mathcal{D}$ and all $\lambda, \lambda' \in \Delta(A)$, we say that **$\boldsymbol{\lambda}$ (first-order) stochastically dominates $\boldsymbol{\lambda'}$ for $\boldsymbol{P}$**, written $\boldsymbol{\lambda\, R^{sd}\, \lambda'}$, if for each upper contour set $U \subseteq A$ of P, $\sum_{a \in U} \lambda_a \geq \sum_{a \in U} \lambda'_a$. If, in addition, some of the latter inequalities are strict for some upper contour

sets, $\boldsymbol{\lambda}$ **strictly stochastically dominates** $\boldsymbol{\lambda'}$ **for** $\boldsymbol{P}$ and we write $\boldsymbol{\lambda\, P^{sd}\, \lambda'}$.

In the context of random rules, strategy-proofness says that the lottery the agent gets with a true report should stochastically dominate any other lottery he gets with a lie.

Strategy-proofness: For all $P, P' \in \mathcal{D}$, $\varphi(P)\, R^{sd}\, \varphi(P')$.

Local strategy-proofness is the same as in the deterministic model, except that comparison of lotteries involves stochastic dominance.

Local Strategy-proofness: For all $P, P' \in \mathcal{D}$ with $(P, P') \in \mathcal{E}$, $\varphi(P)\, R^{sd}\, \varphi(P')$.

Next, we define LGE for random rules.

Definition 7. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies the **random local-global equivalence (RLGE)** if each locally strategy-proof random rule on $\mathcal{D}$ is strategy-proof.

To verify RLGE, one has to work with all random rules, including deterministic ones. Therefore, RLGE is more demanding than LGE and conditions characterizing RLGE will be stronger than strong connectedness or Property DL. To state a condition necessary for RLGE, let us define the following notation. For all non-empty $B, C \subseteq A$ with $|B| \geq 2$ and $C \subsetneq B$, let $\boldsymbol{\mathcal{D}(C, B)} \equiv \{P \in \mathcal{D} : C\, P\, (B \backslash C)\}$ be the set of types that prefer each alternative in C to each alternative in $B \backslash C$ (i.e., when restricted to B, P has C as an upper contour set). Now we define a property that strengthens strong connectedness.

Definition 8. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ is **hyper-connected** if for all non-empty $B, C \subseteq A$ with $|B| \geq 2$ and $C \subsetneq B$, $\langle \mathcal{D}(C, B), \mathcal{E}|_{\mathcal{D}(C,B)} \rangle$ is connected.

Clearly, hyper-connectedness is stronger than strong connectedness because the latter restricts the set C in the definition of hyper-connectedness to singletons. Then how is it related to the no-restoration property by Sato (2013a), another condition stronger than strong connectedness? A simple argument shows that hyper-connectedness is weaker than the no-restoration property.

$$P_1 \equiv abcd \xleftrightarrow{\{\{a,b\}\}} P_2 \equiv bacd \xleftrightarrow{\{\{a,d\},\{b,d\},\{c,d\}\}} P_3 \equiv dbac$$

Figure 7: An environment satisfying the no-restoration property but not RLGE (Example 8).

Fact 1. *The no-restoration property implies hyper-connectedness, which implies strong connectedness.*[25]

Our next result shows that hyper-connectedness is a necessary condition for RLGE.

Theorem 2. *If an environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies RLGE, then it is hyper-connected.*

Let us turn to sufficiency for RLGE. Does hyper-connectedness imply RLGE? As it turns out, the latter implication does not hold. In fact, even the no-restoration property, a stronger condition, is not sufficient for RLGE and neither is it necessary. The next two examples establish this point.

Example 8 (*The no-restoration property—hence, hyper-connectedness—does not imply RLGE*)**.** Assume that $A = \{a, b, c, d\}$ and consider a symmetry environment $\langle \mathcal{D}, \mathcal{E}^{\text{sym}} \rangle$ in Figure 7, where $\mathcal{D} \equiv \{P_1, P_2, P_3\}$. Around each edge $(P, P') \in \mathcal{E}^{\text{sym}}$, we indicate the set $P \triangle P'$ of pairs of alternatives over which P and P' disagree. It is simple to check that $\langle \mathcal{D}, \mathcal{E}^{\text{sym}} \rangle$ satisfies the no-restoration property. Now define a rule φ as $\varphi(P_1) = \varphi(P_2) = \frac{1}{2}b + \frac{1}{2}c$ and $\varphi(P_3) = \frac{1}{2}a + \frac{1}{2}d$. Then φ is locally strategy-proof but is not strategy-proof because $\varphi(P_1)\, R_1^{sd}\, \varphi(P_3)$ does not hold. Thus, RLGE fails. $\triangle$

Example 9 (*RLGE does not imply the no-restoration property*)**.** Assume that $A = \{a, b, c, d\}$ and consider an (asymmetry) environment $\langle \mathcal{D}, \mathcal{E} \rangle$ in Figure 8, where $\mathcal{D} \equiv \{P_1, P_2, P_3, P_4, P_5\}$. One can show that $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies RLGE (by, for example, appealing to Property ULGO, which we introduce below). However, it

[25] To prove that the no-restoration property implies hyper-connectedness, let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment satisfying the no-restoration property. Let $B, C \subseteq A$ be non-empty sets with $|B| \geq 2$ and $C \subsetneq B$. Let $P, P' \in \mathcal{D}(C, B)$. By the no-restoration property, there is $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$ such that for all $a, b \in A$, π is without $\{a, b\}$-restoration. Since $P, P' \in \mathcal{D}(C, B)$, this implies that each $P_j \in \{P_1, \cdots, P_k\}$ satisfies $C\, P_j\, (B \backslash C)$, so that $P_j \in \mathcal{D}(C, B)$. That is, $\pi \in \Pi(P, P', \mathcal{E}|_{\mathcal{D}(C,B)})$.

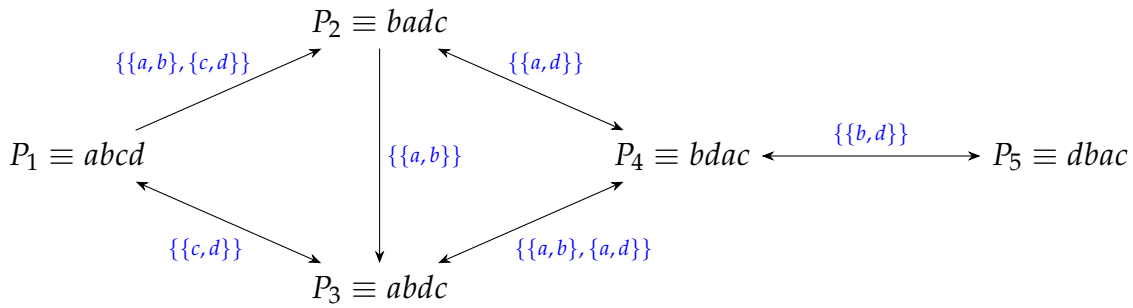

Figure 8: An environment satisfying RLGE but not the no-restoration property (Example 9).

violates the no-restoration property. To see this, consider $P_2, P_3 \in \mathcal{D}$. There are only two paths from P_3 to P_2, namely (P_3, P_1, P_2) and (P_3, P_4, P_2). The former is with $\{c, d\}$-restoration and the latter is with $\{a, d\}$-restoration. $\triangle$

There are several ways of "fixing" P_3 in Example 8 to restore RLGE. For instance, we may replace P_3 by $P_3' \equiv bdac$ and obtain $\varphi_a(P_1) \geq \varphi_a(P_2) \geq \varphi_a(P_3')$ (which is needed for $\varphi(P_1)\, R_1^{sd}\, \varphi(P_3')$) for a locally strategy-proof rule φ. In going from P_2 to P_3', a still becomes less desirable. However, since $\varphi_b(P_2) = \varphi_b(P_3')$, local strategy-proofness implies $\varphi_a(P_2) \geq \varphi_a(P_3')$. The key is that while alternatives are reshuffled in going from P_2 to P_3', $\{a, b, c, d\}$ is partition into $\{b\}$ and $\{a, c, d\}$, both P_2 and P_3' prefer $\{b\}$ to $\{a, c, d\}$, and P_2 places a at the top in the second cell $\{a, c, d\}$. Alternatively, we may replace P_3 by $P_3'' \equiv dbca$ (or $dcba$). Although $\{a, b, c, d\}$ is not divided further in this case and P_2 does not rank a at the top in the cell (i.e., $\{a, b, c, d\}$) to which a belongs, P_3'' ranks a at the bottom in the cell to which a belongs. By local strategy-proofness applied to P_3'' and P_2, it follows that $\varphi_a(P_3'') \leq \varphi_a(P_2)$. In fact, one can formulate similar arguments for other upper contour sets and other types and show that the two environments involving P_3' and P_3'' satisfy RLGE. However, these types of arguments do not apply to Example 8. Consequently, a locally strategy-proof rule φ has the flexibility to behave arbitrarily, possibly assigning a sufficiently high probability to a at P_3, in violation of $\varphi_a(P_1) \geq \varphi_a(P_3)$ as demanded by strategy-proofness.

This observation hints at some elements that should be captured by a suf-

ficient condition for RLGE: not only should the condition be a strengthening of strong connectedness or Property DL but it should also limit the manner in which alternatives overtake others. To introduce one such property, let us define the following notation and concepts. For $P \in \mathcal{D}$ satisfying, say, $C_1\, P\, C_2\, P\, x\, P\, \{y, z\}$, where $\{C_1, C_2, \{x\}, \{y, z\}\}$ is a partition of A, we simply write $\boldsymbol{P = [C_1, C_2, x, \{y, z\}]}$; similarly, we write the restriction $P|_{\{x,y,z\}}$ of P to a subset $\{x, y, z\}$ as $P|_{\{x,y,z\}} = [x, \{y, z\}]$. For all $P, P' \in \mathcal{D}$, we can always find a partition $\{C_1, C_2, \cdots, C_t\}$ of A such that $P = [C_1, C_2, \cdots, C_t]$ and $P' = [C_1, C_2, \cdots, C_t]$. The **associated partition for $\boldsymbol{P}$ and $\boldsymbol{P'}$** (and for the edge (P, P') if $(P, P') \in \mathcal{E}$) is the *finest* of all partitions $\{C_1, C_2, \cdots, C_t\}$ with the above property. We always write the associated partition $\{C_1, C_2, \cdots, C_t\}$ for P and P' in such an order that P and P' rank C_1 first, C_2 second, and so on. For each $P \in \mathcal{D}$ and each $B \subseteq A$, let $\boldsymbol{L(P, B)} \equiv \{a \in A : \text{for each } b \in B,\, b\, P\, a\}$ be the lower contour set of P at B. Clearly, if B is an upper contour set of P, then $L(P, B) = A \backslash B$.

Definition 9. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies the **upper-lower no-restoration property with limited overtaking**, or simply **Property ULLO**, if for all $P, P' \in \mathcal{D}$ and each upper contour set $B \subseteq A$ of P, there is a path $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$ such that (i) for each $b \in B$ and each $c \in L(P, B) = A \backslash B$, π is without $\{b, c\}$-restoration; and (ii) for each $\ell \in \{1, \cdots, k-1\}$, letting $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ be the associated partition for $(P_\ell, P_{\ell+1})$, for each $s \in \{1, \cdots, t_\ell\}$, $P_\ell|_{C_s^\ell} = [C_s^\ell \cap B, C_s^\ell \backslash B]$ or $P_{\ell+1}|_{C_s^\ell} = [C_s^\ell \backslash B, C_s^\ell \cap B]$.[26]

Property ULLO consists of two conditions. Condition (i) stipulates no-restoration between upper and lower contour sets. Upper (and lower) contour sets come into play because preferences based on stochastic dominance compare probabilities for upper contour sets. Clearly, condition (i) is intermediate in strength between the no-restoration property and Property DL (and hence strong connectedness). On the other hand, condition (ii), which we call the **limited-overtaking**

[26]Roy and Sadhukhan (2022) provide a condition on the domain, called the *upper-contour no-restoration (UCNR) property*, which ensures that strategy-proofness is equivalent to a combination of weaker properties. While the question they explore is different from ours, the UCNR property has condition (i) of Property ULLO as one of its requirements. Cho and Park (2025) discuss the relationship in detail and apply our results to Roy and Sadhukhan (2022).

condition, constrains how alternatives in the upper contour set B are overtaken by alternatives in the lower contour set $A \backslash B$. As B is overtaken along the identified path π, P_ℓ should rank its subsets, $C_s^\ell \cap B$, at the top in the cell to which it belongs, and if not, $P_{\ell+1}$ should rank $C_s^\ell \cap B$ at the bottom in the cell. If an environment in question is an adjacency environment, condition (i) implies the limited-overtaking condition trivially: at each edge in a path, there is exactly one pair $(b, c) \in A^2$ such that b is overtaken by c; and for such $\{b, c\}$, whenever $\{b, c\} \cap B \neq \emptyset$ and $\{b, c\} \backslash B \neq \emptyset$, we have $P_\ell|_{\{b,c\}} = [b, c]$ and $P_{\ell+1}|_{\{b,c\}} = [c, b]$.

Example 10 (*Environments satisfying Property ULLO*)**.** Property ULLO is satisfied by several familiar domains in the literature if appropriate sets of edges, such as $\mathcal{E}^{\text{adj}}$, are appended to them. Examples include (i) the universal, single-peaked, single-dipped, single-crossing, and hybrid domains when alternatives are abstract and have no structure; and (ii) some lexicographically separable domains and the domain under committee formation when alternatives are multi-dimensional. $\triangle$

Our next main result says that Property ULLO suffices for RLGE.

Theorem 3. *If an environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies Property ULLO, then it satisfies RLGE.*

A simple corollary obtains when we restrict the above theorem to adjacency environments.

Corollary 3 (Cho, 2016)**.** *Let $\langle \mathcal{D}, \mathcal{E}^{adj} \rangle$ be an adjacency environment. If $\langle \mathcal{D}, \mathcal{E}^{adj} \rangle$ satisfies the no-restoration property (or condition (i) of Property ULLO), then it satisfies RLGE.*

Now we introduce another sufficient condition for RLGE that is stronger but simpler to check than Property ULLO. In adjacency environments, for any edge $(P, P') \in \mathcal{E}$, the associated partition for (P, P') is particularly simple: $\{\{c_1\}, \cdots, \{c_{s-1}\}, \{x, y\}, \{c_{s+1}\}, \cdots, \{c_t\}\}$; i.e., one cell is a doubleton and all the others are singletons. With only a small number of alternatives in each cell, a locally strategy-proof rule has little room in which to behave differently from

strategy-proof rules. Then does this observation on adjacency environments generalize? What if each cell has at most three alternatives or at most four? Our next property limits each cell to have at most three alternatives.

Definition 10. An environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies the **upper-lower no-restoration property with gradual overtaking**, or simply **Property ULGO**, if for all $P, P' \in \mathcal{D}$ and each upper contour set $B \subseteq A$ of P, there is a path $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$ such that (i) for each $b \in B$ and each $c \in L(P, B) = A \backslash B$, π is without $\{b, c\}$-restoration; and (ii) for each $\ell \in \{1, \cdots, k-1\}$, letting $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ be the associated partition for $(P_\ell, P_{\ell+1})$, for each $s \in \{1, \cdots, t_\ell\}$, $|C_s^\ell| \leq 3$.

Property ULGO differs from Property ULLO only in condition (ii). If each cell has at most three alternatives as condition (ii) requires, then each alternative can overtake at most two alternatives at each edge along the path π (the same would be true if A has at most three alternatives in the first place). That is, overtaking along π is "gradual". While the limited-overtaking condition of Property ULLO also constrains the overtaking behavior, its focus is hardly on the gradual change in preferences. Nonetheless, our next result shows that condition (ii) of Property ULGO is a stronger form of the limited-overtaking condition.

Proposition 5. *Property ULGO implies Property ULLO.*

An immediate corollary to the above proposition is that Property ULGO implies RLGE. Though stronger than Property ULLO, Property ULGO is a more convenient test for RLGE. Condition (ii) of Property ULGO is not contingent on the upper contour set B chosen (unlike the limited-overtaking condition of Property ULLO). Therefore, once a candidate path satisfying condition (i) is identified, we can just check the cardinality of cells in the associated partitions.

It is noteworthy that one cannot extend the spirit of Property ULGO further without compromising RLGE. That is, if some associated partition has a cell containing four alternatives or more, RLGE is no longer guaranteed. Even though the environment of Example 8 satisfies the no-restoration property, the associated partition for (P_2, P_3) is $\{\{a, b, c, d\}\}$ and RLGE fails.

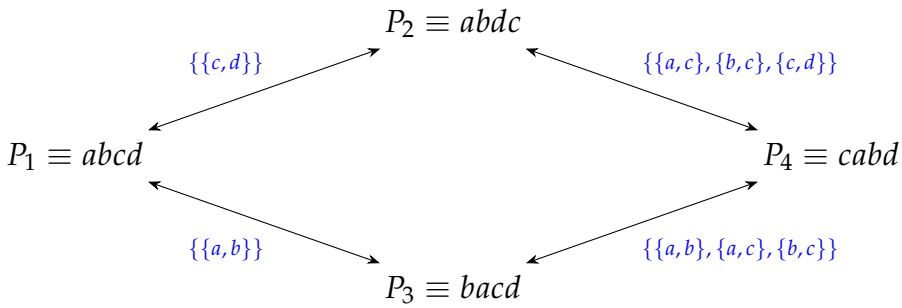

Figure 9: An environment satisfying hyper-connectedness but failing Property ULLO (condition (i)) and RLGE (Example 11).

Finally, we conclude this section by discussing the gap between hyper-connectedness and Property ULLO. We already know that strong connectedness, Property DL, and LGE are equivalent. Given that hyper-connectedness strengthens strong connectedness and Property ULLO similarly strengthens Property DL, one may wonder whether hyper-connectedness implies Property ULLO (or at least its condition (i)). Our next example answers the question in the negative.

Example 11 (*Hyper-connectedness does not imply condition (i) of Property ULLO.*)**.** Assume that $A = \{a, b, c, d\}$ and consider a symmetry environment $\langle \mathcal{D}, \mathcal{E}^{\text{sym}} \rangle$ in Figure 9, where $\mathcal{D} \equiv \{P_1, P_2, P_3, P_4\}$. One can verify that $\langle \mathcal{D}, \mathcal{E}^{\text{sym}} \rangle$ is hyper-connected. However, it violates condition (i) of Property ULLO. Consider $P_4, P_1 \in \mathcal{D}$ and an upper contour set $B \equiv \{c, a\}$ of P_4. There are only two paths from P_4 to P_1, namely, $\pi \equiv (P_4, P_2, P_1)$ and $\pi' \equiv (P_4, P_3, P_1)$. Yet π is with $\{c, d\}$-restoration and π' is with $\{a, b\}$-restoration, violating condition (i) of Property ULLO. $\triangle$

5 Concluding Remarks

In a model where the agent faces an arbitrary manipulation constraint, we studied the structure of environments for which LGE holds. We provided two conditions, strong connectedness and Property DL, each of which is sufficient and necessary for LGE. The intuition obtained from the characterization guided our search for conditions describing the boundary of RLGE. We showed that hyper-connectedness, a strengthening of strong connectedness, is necessary for RLGE;

and that Property ULLO, a strengthening of Property DL, is sufficient for RLGE. In earlier studies on LGE, the no-restoration perspective, which requires some form of no-restoration as in Property L, has played an important role. Our findings complemented it with the connectedness perspective: one may decompose a given environment into multiple subenvironments and check their connectedness. We established that the connectedness and no-restoration perspectives have a common root if they are refined appropriately.

Finally, we conclude by commenting on remaining issues. For examples of environments satisfying our main conditions, we may turn to some well-known preference domains in the literature and pair them with appropriate sets of edges. Let $\mathcal{D}$ be one of the following domains: (i) the hybrid domains (Chatterji et al., 2022), which include the universal domain (Sato, 2013a) and the single-peaked domain (Sato, 2013a); (ii) the single-dipped domain (Cho, 2016); and (iii) the successive single-crossing domains (Roy and Sadhukhan, 2022), which include the maximal single-crossing domains. Then $\langle \mathcal{D}, \mathcal{E}^{\text{adj}} \rangle$ satisfies Property ULLO and, hence, RLGE.

The strength of our approach and results becomes more evident when they are applied to questions that cannot be answered by existing studies. Our companion paper, Cho and Park (2025), provides such applications, illustrating how to use directed networks, strong connectedness, hyper-connectedness, and Property ULLO to address more concrete questions.

A Appendix: Proofs

A.1 Proof of Proposition 1

Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment satisfying strong connectedness. Let $P, P' \in \mathcal{D}$. We distinguish two cases.

Case 1: *P' is not the inverse of P.*

There exists a pair $(a, b) \in A^2$ such that $a\,P\,b$ and $a\,P'\,b$. Then $P, P' \in \mathcal{D}(a, \{a, b\})$. By strong connectedness, $\langle \mathcal{D}(a, \{a, b\}), \mathcal{E}|_{\mathcal{D}(a,\{a,b\})} \rangle$ is connected,

so that $\Pi(P, P', \mathcal{E}|_{\mathcal{D}(a,\{a,b\})}) \neq \emptyset$ and hence $\Pi(P, P') \neq \emptyset$.

Case 2: *P' is the inverse of P.*

That is, for each $(a, b) \in A^2$, $a\, P\, b$ if and only if $b\, P'\, a$. Fix any pair $(a, b) \in A^2$. Without loss of generality, assume that $a\, P\, b$ and $b\, P'\, a$.

First, suppose that $|\mathcal{D}(b, \{a, b\})| \geq 2$. Let $P'' \in \mathcal{D}(b, \{a, b\}) \backslash \{P'\}$. Since P'' is not the inverse of P, there is a pair $(c, d) \in A^2$ such that $P, P'' \in \mathcal{D}(c, \{c, d\})$. By strong connectedness, $\langle \mathcal{D}(c, \{c, d\}), \mathcal{E}|_{\mathcal{D}(c,\{c,d\})} \rangle$ is connected, so that there is a path $(P_1, \cdots, P_k) \in \Pi(P, P'', \mathcal{E}|_{\mathcal{D}(c,\{c,d\})})$ from P to P''. Again by strong connectedness, $\langle \mathcal{D}(b, \{a, b\}), \mathcal{E}|_{\mathcal{D}(b,\{a,b\})} \rangle$ is connected, so that there is a path $(P_k, \cdots, P_m) \in \Pi(P'', P', \mathcal{E}|_{\mathcal{D}(b,\{a,b\})})$ from P'' to P'. Concatenating the above two paths, we obtain a path $(P_1, \cdots, P_k, \cdots, P_m) \in \Pi(P, P')$.

Second, suppose that $|\mathcal{D}(b, \{a, b\})| = 1$; i.e., $\mathcal{D}(b, \{a, b\}) = \{P'\}$. Since $|\mathcal{D}| \geq 3$, there is $P'' \in \mathcal{D}(a, \{a, b\}) \backslash \{P\}$. By strong connectedness, $\langle \mathcal{D}(a, \{a, b\}), \mathcal{E}|_{\mathcal{D}(a,\{a,b\})} \rangle$ is connected, so that there is a path $(P_1, \cdots, P_k) \in \Pi(P, P'', \mathcal{E}|_{\mathcal{D}(a,\{a,b\})})$ from P to P''. Since P'' is not the inverse of P', there is a pair $(c, d) \in A^2$ such that $P', P'' \in \mathcal{D}(c, \{c, d\})$. Again by strong connectedness, $\langle \mathcal{D}(c, \{c, d\}), \mathcal{E}|_{\mathcal{D}(c,\{c,d\})} \rangle$ is connected, so that there is a path $(P_k, \cdots, P_m) \in \Pi(P'', P', \mathcal{E}|_{\mathcal{D}(c,\{c,d\})})$ from P'' to P'. Concatenating the above two paths, we obtain a path $(P_1, \cdots, P_k, \cdots, P_m) \in \Pi(P, P')$.

A.2 Proof of Proposition 2

Part (1): Let $P \in \mathcal{D}(a, \{a, b\})$ and $P' \in \mathcal{D}(b, \{a, b\})$. By Proposition 1, there is a path $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$. Let $h \in \{1, \cdots, k-1\}$ be the largest integer satisfying $a\, P_h\, b$. By strong connectedness, $\langle \mathcal{D}(a, \{a, b\}), \mathcal{E}|_{\mathcal{D}(a,\{a,b\})} \rangle$ is connected. Thus, there is a path $\hat{\pi} \equiv (\hat{P}_1, \cdots, \hat{P}_j) \in \Pi(P, P_h, \mathcal{E}|_{\mathcal{D}(a,\{a,b\})})$. By our choice of h, each type in the subpath $(P_{h+1}, \cdots, P_k)$ prefers b to a. Then $\hat{\pi} \cup \pi|_{[P_h, P_k]} = (\hat{P}_1, \cdots, \hat{P}_j, P_{h+1}, \cdots, P_k)$ is a path from P to P' without $\{a, b\}$-restoration.

Part (2): By part (1) of this proposition, it suffices to note that $\mathcal{D}(a, B) \subseteq \mathcal{D}(a, \{a, b\})$ and $\mathcal{D}(b, B) \subseteq \mathcal{D}(b, \{a, b\})$.

A.3 Proof of Theorem 1

First, for all $P, P' \in \mathcal{D}$ and each $x \in A$, define $\boldsymbol{L(P, P', x)} \equiv L(P, x) \cap L(P', x)$.

To prove Theorem 1, below we show that (1) implies (2); that (2) implies (3); and that (3) implies (1).

A.3.1 Showing that (1) implies (2)

Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment satisfying LGE. Suppose, by contradiction, that for some $B \subseteq A$ with $|B| \geq 2$ and $a \in B$, $\langle \mathcal{D}(a, B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$ is not connected. Find $P^*, P^{**} \in \mathcal{D}(a, B)$ such that $\Pi(P^*, P^{**}, \mathcal{E}|_{\mathcal{D}(a,B)}) = \emptyset$. Define a rule φ as for each $P \in \mathcal{D}$,

(i) if $P \notin \mathcal{D}(a, B)$, then $\varphi(P) = \max_P B \ (\neq a)$;

(ii) if $P \in \mathcal{D}(a, B)$ and $\Pi(P^*, P, \mathcal{E}|_{\mathcal{D}(a,B)}) \neq \emptyset$, then $\varphi(P) = \max_P B \backslash \{a\}$[27]; and

(iii) if $P \in \mathcal{D}(a, B)$ and $\Pi(P^*, P, \mathcal{E}|_{\mathcal{D}(a,B)}) = \emptyset$, then $\varphi(P) = \max_P B = a$.

We show that φ is locally strategy-proof. Only those types to which (ii) applies will potentially misrepresent, to get a. However, local misrepresentations cannot give a to them because there is no edge from a type to which (ii) applies to a type to which (iii) applies.

However, φ is not strategy-proof since $a = \varphi(P^{**}) \, P^* \, \varphi(P^*) \neq a$. Thus, LGE fails, a contradiction.

A.3.2 Showing that (2) implies (3)

Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment satisfying strong connectedness. We prove that Property DL follows by the induction on the cardinality of the set $L(P, P', a)$. Let $n \equiv |A|$. First, for all $P, P' \in \mathcal{D}$ and each $a \in A$ with $|L(P, P', a)| = n - 1$, there exists $\pi \in \Pi(P, P')$ such that for each $b \in L(P, a)$, π is without $\{a, b\}$-restoration. This is because $P, P' \in \mathcal{D}(a, A)$ and by strong connectedness, $\langle \mathcal{D}(a, A), \mathcal{E}|_{\mathcal{D}(a,A)} \rangle$ is connected.

[27] In an abuse of notation, we assume that (ii) applies to P^*, so that $\varphi(P^*) \neq a$.

Next, let $m \in \{0, 1, \cdots, n-2\}$ and assume that the following **Induction Hypothesis** holds: for all $P, P' \in \mathcal{D}$ and each $a \in A$ with $|L(P, P', a)| \geq m + 1$, there is $\pi \in \Pi(P, P')$ such that for each $b \in L(P, a)$, π is without $\{a, b\}$-restoration. We now show that a statement similar to the Induction Hypothesis follows even if $m + 1$ is replaced by m.

Let $P, P' \in \mathcal{D}$ and $a \in A$ satisfy $|L(P, P', a)| = m$. Write $L(P, P', a) = \{c_1, \cdots, c_m\}$ (if $m = 0$, then this set is empty and some arguments below become vacuous). We distinguish two cases.

Case 1: $L(P, a) \backslash L(P, P', a) \neq \emptyset$.

Write $L(P, a) \backslash L(P, P', a) = \{b_1, \cdots, b_s\}$ (where $s \geq 1$), so that $L(P, a) = \{b_1, \cdots, b_s, c_1, \cdots, c_m\}$. Since $\{b_1, \cdots, b_s\}\, P'\, a\, P'\, \{c_1, \cdots, c_m\}$, we need to construct a path from P to P' having:

(i) for each $\ell \in \{1, \cdots, s\}$, one crossing of the a-b_ℓ border; and

(ii) for each $\ell \in \{1, \cdots, m\}$, no crossing of the a-c_ℓ border.

Let $B \equiv \{a\} \cup L(P, P', a) = \{a, c_1, \cdots, c_m\}$. First, $P, P' \in \mathcal{D}(a, B)$. By strong connectedness, $\langle \mathcal{D}(a, B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$ is connected. Thus, there is $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P', \mathcal{E}|_{\mathcal{D}(a,B)})$. Along path π, a is always preferred to $B \backslash \{a\}$. Thus, for each $\ell \in \{1, \cdots, m\}$, π does not cross the a-c_ℓ border.

Now let $h \in \{1, \cdots, k-1\}$ be the largest integer such that for some $\ell \in \{1, \cdots, s\}$, $a\, P_h\, b_\ell$; i.e., P_h is the type in π just before the last crossing of the a-b_1, $\cdots$, a-b_s borders. Let us distinguish two subcases.

Case 1.1: $h = 1$.

Then $a\, P_1\, \{b_1, \cdots, b_s, c_1, \cdots, c_m\}$ and for each $j \in \{2, \cdots k\}$, $\{b_1, \cdots, b_s\}\, P_j\, a\, P_j\, \{c_1, \cdots, c_m\}$. Thus, for each $x \in L(P, a)$, π is without $\{a, x\}$-restoration.

Case 1.2: $h \in \{2, \cdots, k-1\}$.

In this case, P, P_h, and P' are distinct (thus, the assumption $|\mathcal{D}| \geq 3$ is invoked). Suppose, without loss of generality, that P_h satisfies $a\, P_h\, b_\ell$ for each $\ell \in \{t, \cdots, s\}$, where $t \leq s$ (t is the smallest integer with this property). By construction, $\pi|_{[P_h, P']}$ has:

(i) for each $\ell \in \{1, \cdots, t-1\}$, no crossing of the a-b_ℓ border[28];

(ii) for each $\ell \in \{t, \cdots, s\}$, one crossing of the a-b_ℓ border; and

(iii) for each $\ell \in \{1, \cdots, m\}$, no crossing of the a-c_ℓ border.

Consider a triple (P, P_h, a). Note that $L(P, P_h, a) = \{b_t, \cdots, b_s, c_1, \cdots, c_m\}$. Since $|L(P, P_h, a)| \geq m+1$, the Induction Hypothesis implies that there is $\hat{\pi} \in \Pi(P, P_h)$ such that for each $x \in L(P, a)$, $\hat{\pi}$ is without $\{a, x\}$-restoration. For a new path $\hat{\pi} \cup \pi|_{[P_h, P']}$, it follows that for each $x \in L(P, a)$, $\hat{\pi} \cup \pi|_{[P_h, P']}$ is without $\{a, x\}$-restoration, as desired.

Case 2: $L(P, a) = \{c_1, \cdots, c_m\} = L(P, P', a)$.

If $m = 0$, then Property DL requires only the existence of a path from P to P', which follows from strong connectedness of $\langle \mathcal{D}, \mathcal{E} \rangle$ and Proposition 1. Thus, we may assume $m \geq 1$. Let $B \equiv \{a\} \cup L(P, P', a) = \{a, c_1, \cdots, c_m\}$. First, $P, P' \in \mathcal{D}(a, B)$. Strong connectedness implies that $\langle \mathcal{D}(a, B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$ is connected, so that there is $\pi \in \Pi(P, P', \mathcal{E}|_{\mathcal{D}(a,B)})$. Along path π, a is always preferred to $B \backslash \{a\}$. Thus, for each $x \in L(P, a)$, π is without $\{a, x\}$-restoration.

A.3.3 Showing that (3) implies (1)

Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment satisfying Property DL. Suppose, by contradiction, that $\langle \mathcal{D}, \mathcal{E} \rangle$ fails LGE; i.e., there is a rule φ on $\mathcal{D}$ that is locally strategy-proof but is not strategy-proof. Define $\Theta \equiv \left\{ (\hat{P}, \hat{P}') \in \mathcal{D}^2 : \varphi(\hat{P}') \, \hat{P} \, \varphi(\hat{P}) \right\} (\neq \emptyset)$. Choose $(P, P') \in \Theta$ such that $L\left(P, P', \varphi(P')\right)$ is maximal in the collection $\left\{ L\left(\hat{P}, \hat{P}', \varphi(\hat{P}')\right) : (\hat{P}, \hat{P}') \in \Theta \right\}$ in the set inclusion sense.[29] Let $a_1 \equiv \varphi(P')$. Figure 10 illustrates our argument. In the figure, the alternative that φ chooses for each type is in parentheses next to that type and is highlighted in **red**.

By Property DL, there is a path $\pi_1 \equiv (P_{1,1}, \cdots, P_{1,k_1}) \in \Pi(P, P')$ such that for each $x \in L(P, a_1)$, π_1 is without $\{a_1, x\}$-restoration. Since φ is locally strategy-proof, $(P, P') \notin \mathcal{E}$, so that $k_1 \geq 3$ (thus, the assumption $|\mathcal{D}| \geq 3$ is invoked). Let $h_1 \in \{1, \cdots, k_1\}$ be the smallest number with $\varphi(P_{1,h_1}) = a_1$.[30]

[28] If $t = 1$, (i) does not apply.

[29] That is, there is no $(\hat{P}, \hat{P}') \in \Theta$ with $L\left(\hat{P}, \hat{P}', \varphi(\hat{P}')\right) \supsetneq L\left(P, P', \varphi(P')\right)$.

[30] Note that $h_1 \neq 1$ because $P_{1,1} = P$ and $a_1 \, P \, \varphi(P)$.

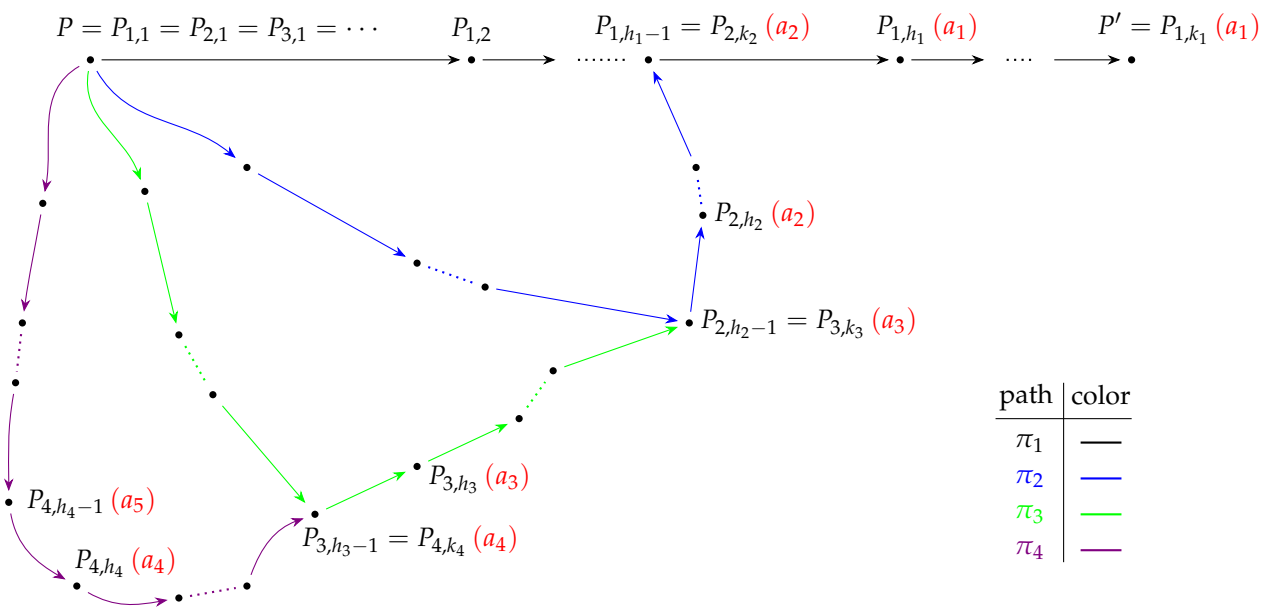

Figure 10: Illustrating the proof that (3) implies (1)

By construction, $a_1 \neq a_2 \equiv \varphi(P_{1,h_1-1})$. Note that $a_2 \, P_{1,h_1-1} \, a_1$ (by local strategy-proofness). Now we distinguish two cases.

Case 1: $a_2 \, P_{1,h_1} \, a_1$.

The assumption for the present case means that strategy-proofness is violated for P_{1,h_1}.

Now we show that $L(P_{1,1}, P_{1,k_1}, a_1) \subsetneq L(P_{1,h_1}, P_{1,h_1-1}, a_2)$, which contradicts maximality of $L(P_{1,1,}, P_{1,k_1}, a_1)$. Let $x \in LHS$; i.e., $a_1 \, P_{1,1} \, x$ and $a_1 \, P_{1,k_1} \, x$. Since π_1 is without $\{a_1, x\}$-restoration, $a_1 \, P_{1,h_1-1} \, x$ and $a_1 \, P_{1,h_1} \, x$, i.e., $x \in RHS$. Further, the inclusion is strict because $a_1 \in RHS$ (by the assumption for the present case and $a_2 \, P_{1,h_1-1} \, a_1$) but $a_1 \notin LHS$.

Case 2: $a_1 \, P_{1,h_1} \, a_2$.

The assumption for the present case implies that $a_2 \, P \, a_1$ and $a_2 \, P_{1,h_1-1} \, a_1$.[31]

By Property DL, there is a path $\pi_2 \equiv (P_{2,1}, \cdots, P_{2,k_2}) \in \Pi(P, P_{1,h_1-1})$ such that for each $x \in L(P, a_2)$, π_2 is without $\{a_2, x\}$-restoration. Since φ is locally strategy-proof, $(P, P_{1,h_1-1}) \notin \mathcal{E}$, so that $k_2 \geq 3$. Let $h_2 \in \{1, \cdots, k_2\}$ be the

[31] To see $a_2 \, P \, a_1$, suppose not. Then $a_2 \in L(P_{1,1}, a_1)$ and π_1 is with $\{a_1, a_2\}$-restoration, a contradiction.

smallest number with $\varphi(P_{2,h_2}) = a_2$.[32] By construction, $a_2 \neq a_3 \equiv \varphi(P_{2,h_2-1})$.

Note that $a_3 \, P_{2,h_2-1} \, a_2$ (by local strategy-proofness) and $a_2 \, P_{2,h_2-1} \, a_1$ (because π_2 is without $\{a_2, a_1\}$-restoration), so that $a_3 \, P_{2,h_2-1} \, a_2 \, P_{2,h_2-1} \, a_1$. In particular, a_3, a_2, and a_1 are all distinct. So far, we have chosen four distinct types from $\mathcal{D}$, namely P, P', P_{1,h_1-1}, and P_{2,h_2-1}. If $|\mathcal{D}| = 3$, this is a contradiction. Thus, assume, henceforth, that $|\mathcal{D}| \geq 4$. We distinguish two cases.

Case 2.1: $a_3 \, P_{2,h_2} \, a_2$.

The assumption for the present case means that strategy-proofness is violated for P_{2,h_2}.

Note that $(P_{2,1}, P_{2,k_2}) \in \Theta$. Now we show that $L(P_{1,1}, P_{1,k_1}, a_1) \subsetneq L(P_{2,1}, P_{2,k_2}, a_2)$, which contradicts maximality of $L(P_{1,1}, P_{1,k_1}, a_1)$. (Recall $P_{2,1} = P$ and $P_{2,k_2} = P_{1,h_1-1}$.) Let $x \in LHS$; i.e., $a_1 \, P_{1,1} \, x$ and $a_1 \, P_{1,k_1} \, x$. Since $a_2 \, P_{1,1} \, a_1$, we have $a_2 \, P_{1,1} \, a_1 \, P_{1,1} \, x$. Since $a_1 \, P_{1,1} \, x$, $a_1 \, P_{1,k_1} \, x$, and π_1 is without $\{a_1, x\}$-restoration, we have $a_1 \, P_{1,h_1-1} \, x$. Together with $a_2 \, P_{1,h_1-1} \, a_1$, we have $a_2 \, P_{1,h_1-1} \, a_1 \, P_{1,h_1-1} \, x$. Thus, $x \in RHS$. Further, the inclusion is strict because $a_1 \in RHS$ but $a_1 \notin LHS$.

Case 2.2: $a_2 \, P_{2,h_2} \, a_3$.

The assumption for the present case implies that $a_3 \, P \, a_2 \, P \, a_1$ and $a_3 \, P_{2,h_2-1} \, a_2 \, P_{2,h_2-1} \, a_1$.[33]

By Property DL, there is a path $\pi_3 \equiv (P_{3,1}, \cdots, P_{3,k_3}) \in \Pi(P, P_{2,h_2-1})$ such that for each $x \in L(P, a_3)$, π_3 is without $\{a_3, x\}$-restoration. Since φ is locally strategy-proof, $(P, P_{2,h_2-1}) \notin \mathcal{E}$, so that $k_3 \geq 3$. Let $h_3 \in \{1, \cdots, k_3\}$ be the smallest number with $\varphi(P_{3,h_3}) = a_3$.[34] By construction, $a_3 \neq a_4 \equiv \varphi(P_{3,h_3-1})$.

Note that $a_4 \, P_{3,h_3-1} \, a_3$ (by local strategy-proofness) and P_{3,h_3-1} most prefers a_3 in $\{a_3, a_2, a_1\}$ (because both P and P_{2,h_2-1} most prefer a_3 in $\{a_3, a_2, a_1\}$ and π_3 is without $\{a_3, a_2\}$- and $\{a_3, a_1\}$-restoration), so that $a_4 \, P_{3,h_3-1} \, a_3 \, P_{3,h_3-1} \, \{a_2, a_1\}$. In particular, a_4, a_3, a_2, a_1 are all distinct. If $|A| = 3$, this is a contradiction. Further, since we have so far chosen five distinct types from $\mathcal{D}$, namely $P, P', P_{1,h_1-1}, P_{2,h_2=1}$, and P_{3,h_3-1}, another contradiction arises if $|\mathcal{D}| = 4$. Thus, assume, henceforth,

[32] Note that $h_2 \neq 1$ because $P_{2,1} = P$, $a_2 \, P \, a_1$, and $a_1 \, P \, \varphi(P)$.

[33] To see $a_3 \, P \, a_2 \, P \, a_1$, we already know $a_2 \, P \, a_1$. To get $a_3 \, P \, a_2$, suppose not; i.e., $a_3 \in L(P, a_2)$. Then $a_2 \, P_{2,1} \, a_3$, $a_3 \, P_{2,h_2-1} \, a_2$, and $a_2 \, P_{2,h_2} \, a_3$, violating that π_2 is without $\{a_2, a_3\}$-restoration, a contradiction.

[34] Note that $h_3 \neq 1$ because $P_{3,1} = P$, $a_3 \, P \, a_2 \, P \, a_1$, and $a_1 \, P \, \varphi(P)$.

that $|A| \geq 4$ and $|\mathcal{D}| \geq 5$. We distinguish two cases.

Case 2.2.1: $a_4 \, P_{3,h_3} \, a_3$.

The assumption for the present case means that strategy-proofness is violated for P_{3,h_3}.

Note that $(P_{3,1}, P_{3,k_3}) \in \Theta$. Now we show that $L(P_{1,1}, P_{1,k_1}, a_1) \subsetneq L(P_{3,1}, P_{3,k_3}, a_3)$, which contradicts maximality of $L(P_{1,1}, P_{1,k_1}, a_1)$. (Recall $P_{3,1} = P$ and $P_{3,k_3} = P_{2,h_2-1}$.) Let $x \in LHS$; i.e., $a_1 \, P_{1,1} \, x$ and $a_1 \, P_{1,k_1} \, x$. Since $a_3 \, P_{1,1} \, a_1$, we have $a_3 \, P_{1,1} \, x$, where $P_{1,1} = P_{3,1}$. To get $a_3 \, P_{3,k_3} \, x$, we first show $a_2 \, P_{2,k_2} \, x$. Since $a_1 \, P_{1,1} \, x$, $a_1 \, P_{1,k_1} \, x$, and π_1 is without $\{a_1, x\}$-restoration, we have $a_1 \, P_{1,h_1-1} \, x$. This, together with $a_2 \, P_{1,h_1-1} \, a_1$, implies $a_2 \, P_{1,h_1-1} \, x$, where $P_{1,h_1-1} = P_{2,k_2}$. Now since $a_2 \, P_{1,1} \, x$, $a_2 \, P_{2,k_2} \, x$, and π_2 is without $\{a_2, x\}$-restoration, we have $a_2 \, P_{2,h_2-1} \, x$. Together with $a_3 \, P_{2,h_2-1} \, a_2$, it follows that $a_3 \, P_{2,h_2-1} \, x$, where $P_{2,h_2-1} = P_{3,k_3}$. Thus, $x \in RHS$. Further, the inclusion is strict because $a_2 \in RHS$ (since $a_3 \, P \, a_2 \, P \, a_1$ and $a_3 \, P_{2,h_2-1} \, a_2 \, P_{2,h_2-1} \, a_1$) but $a_2 \notin LHS$.

Case 2.2.2: $a_3 \, P_{3,h_3} \, a_4$.

By the assumption for the present case, $a_4 \, P \, a_3 \, P \, a_2 \, P \, a_1$ and $a_4 \, P_{3,h_3-1} \, a_3 \, P_{3,h_3-1} \, \{a_2, a_1\}$.[35]

By Property DL, there is a path $\pi_4 \equiv (P_{4,1}, ..., P_{4,k_4}) \in \Pi(P, P_{3,h_3-1})$ such that for each $x \in L(P, a_4)$, π_4 is without $\{a_4, x\}$-restoration. Since φ is locally strategy-proof, $(P, P_{3,h_3-1}) \notin \mathcal{E}$, so that $k_4 \geq 3$. Let $h_4 \in \{1, \cdots, k_4\}$ be the smallest number with $\varphi(P_{4,h_4}) = a_4$.[36] By construction, $a_4 \neq a_5 \equiv \varphi(P_{4,h_4-1})$.

Note that $a_5 \, P_{4,h_4-1} \, a_4$ (by local strategy-proofness) and P_{4,h_4-1} most prefers a_4 in $\{a_4, a_3, a_2, a_1\}$ (because both P and P_{3,h_3-1} most prefer a_4 in $\{a_4, a_3, a_2, a_1\}$ and for each $x \in \{a_3, a_2, a_1\}$, π_4 is without $\{a_4, x\}$-restoration), so that $a_5 \, P_{4,h_4-1} \, a_4$ $P_{4,h_4-1} \, \{a_3, a_2, a_1\}$. In particular, a_5, a_4, a_3, a_2, a_1 are all distinct. If $|A| = 4$, this is a contradiction. Further, since we have so far chosen six distinct types from $\mathcal{D}$, namely $P, P', P_{1,h_1-1}, P_{2,h_2-1}, P_{3,h_3-1}$, and P_{4,h_4-1}, another contradiction arises if $|\mathcal{D}| = 5$. Thus, assume, henceforth, that $|A| \geq 5$ and $|\mathcal{D}| \geq 6$.

[35] To see $a_4 \, P \, a_3 \, P \, a_2 \, P \, a_1$, we already know $a_3 \, P \, a_2 \, P \, a_1$ in the present Case 2.2. To get $a_4 \, P \, a_3$, suppose not; i.e., $a_4 \in L(P, a_3)$. Then $a_3 \, P_{3,1} \, a_4$, $a_4 \, P_{3,h_3-1} \, a_3$, and $a_3 \, P_{3,h_3} \, a_4$, violating that π_3 is without $\{a_3, a_4\}$-restoration, a contradiction.

[36] Note that $h_4 \neq 1$ because $P_{4,1} = P$, $a_4 \, P \, a_3 \, P \, a_2 \, P \, a_1$, and $a_1 \, P \, \varphi(P)$.

Repeating our argument above leads to one of the following contradictions in the end. In some Case 2.2. $\cdots$ .2.1, we obtain a contradiction to maximality of $L(P_{1,1}, P_{1,k_1}, x_1)$. If not, in some Case 2.2. $\cdots$ .2.2, we find another alternative $a_{m+1} \in A$ distinct from $a_1, \cdots, a_m$ and another type $P_{m,h_m-1} \in \mathcal{D}$ distinct from $P, P', P_{1,h_1-1}, P_{2,h_2-1}, \cdots, P_{m-1,h_{m-1}-1}$, so that the number of chosen alternatives (i.e., $m+1$) and the number of chosen types (i.e., $m+2$) eventually exceeds the upper bounds of $|A|$ and $|\mathcal{D}|$.

A.4 Proof of Proposition 3

It suffices to prove the "only if" part. Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment with $\mathcal{D}$ satisfying best-value restriction. Suppose that $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies pairwise connectedness. We need to show that for each $B \subseteq A$ with $|B| \geq 2$ and each $a \in B$, $\langle \mathcal{D}(a, B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$ is connected. By pairwise connectedness, it is enough to prove the latter statement for $B \subseteq A$ with $|B| \geq 3$. Our proof is by the induction on $|B|$.

Let $m \in \{2, \cdots, |A|-1\}$ and assume that the following **Induction Hypothesis** holds: for each $B \subseteq A$ with $|B| \leq m$ and each $a \in B$, $\langle \mathcal{D}(a, B), \mathcal{E}|_{\mathcal{D}(a,B)} \rangle$ is connected. We show that a statement similar to the Induction Hypothesis holds even if m is replaced by $m+1$.

Let $B \subseteq A$ be such that $|B| = m+1$. Write $B = \{b_1, b_2, \cdots, b_{m+1}\}$. Let $P, P' \in \mathcal{D}(b_1, B)$. By the Induction Hypothesis, $\langle \mathcal{D}(b_1, \{b_1, \cdots, b_m\}), \mathcal{E}|_{\mathcal{D}(b_1,\{b_1,\cdots,b_m\})} \rangle$ is connected. Thus, there is $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P', \mathcal{E}|_{\mathcal{D}(b_1,\{b_1,\cdots,b_m\})})$. If for each $\ell \in \{1, \cdots, k\}$, $P_\ell \in \mathcal{D}(b_1, \{b_1, \cdots, b_{m+1}\})$, then we are done. Suppose otherwise. Let $h \in \{2, 3, \cdots, k-1\}$ be the smallest integer satisfying $b_{m+1} \; P_h \; b_1$. By construction, (i) $b_1 \; P_{h-1} \; \{b_2, \cdots, b_{m+1}\}$; and (ii) $b_{m+1} \; P_h \; b_1 \; P_h \; \{b_2, \cdots, b_m\}$. The Induction Hypothesis implies that there is $\pi' = (P'_1, \cdots, P'_{k'}) \in \Pi(P_{h-1}, P_k, \mathcal{E}|_{\mathcal{D}(b_1,\{b_1,b_{m+1}\})})$. Let $\ell \in \{1, \cdots, k'\}$ and consider P'_ℓ. Since $\mathcal{D}$ satisfies best-value restriction, (i) and (ii) above imply that for each $b \in \{b_2, \cdots, b_m\}$, $b_1 \; P'_\ell \; b$ or $b_{m+1} \; P'_\ell \; b$. In either case, $P'_\ell \in \mathcal{D}(b_1, B)$, so that $\pi' \in \Pi(P_{h-1}, P_k, \mathcal{E}|_{\mathcal{D}(b_1,B)})$. Thus, $\pi|_{[P_1,P_{h-1}]} \cup \pi' \in \Pi(P, P', \mathcal{E}|_{\mathcal{D}(b_1,B)})$, which is a desired path.

A.5 Proof of Proposition 4

"Only if" part: Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be a pairwise connected environment. Let $P, P' \in \mathcal{D}$ and $a, b \in A$ be distinct. Without loss of generality, we distinguish two cases.

Assume that $P, P' \in \mathcal{D}(a, \{a, b\})$. Pairwise connectedness implies that there is a path $\pi \in \Pi(P, P', \mathcal{E}|_{\mathcal{D}(a,\{a,b\})})$. By construction, π is without $\{a, b\}$-restoration.

Assume that $P \in \mathcal{D}(a, \{a, b\})$ and $P' \in \mathcal{D}(b, \{a, b\})$. By an argument similar to that in the proof of part (1) of Proposition 2, there is a path $\pi \in \Pi(P, P')$ without $\{a, b\}$-restoration.[37]

"If" part: Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment satisfying the pairwise no-restoration property. Let $a, b \in A$ and $P, P' \in \mathcal{D}(a, \{a, b\})$ be distinct. By the pairwise no-restoration property, there is a path $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$ without $\{a, b\}$-restoration. Since $a\, P\, b$ and $a\, P'\, b$, for each $\ell \in \{1, \cdots, k\}$, $a\, P_\ell\, b$. Thus, $\pi \in \Pi(P, P', \mathcal{E}|_{\mathcal{D}(a,\{a,b\})})$, so that $\langle \mathcal{D}(a, \{a, b\}), \mathcal{E}|_{\mathcal{D}(a,\{a,b\})} \rangle$ is connected.

A.6 Proof of Theorem 2[38]

Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment satisfying RLGE. Suppose, by contradiction, that it is not hyper-connected; i.e., there are $B, C \subseteq A$ with $|B| \geq 2$ and $C \subsetneq B$ such that $\langle \mathcal{D}(C, B), \mathcal{E}|_{\mathcal{D}(C,B)} \rangle$ is not connected. Find $P^*, P^{**} \in \mathcal{D}(C, B)$ such that $\Pi(P^*, P^{**}, \mathcal{E}|_{\mathcal{D}(C,B)}) = \emptyset$. We already know that if C is singleton, by strong connectedness (which is equivalent to LGE), $\langle \mathcal{D}(C, B), \mathcal{E}|_{\mathcal{D}(C,B)} \rangle$ is connected. Assume, henceforth, that $|C| = m \geq 2$. Since $C \subsetneq B$, $|B| \geq m + 1$. Now we proceed in two steps.

Step 1: *Defining a rule φ.*

Define a rule φ as for each $P \in \mathcal{D}$,

(i) if $P \in \mathcal{D}(C, B)$ and $\Pi(P^*, P, \mathcal{E}|_{\mathcal{D}(C,B)}) \neq \emptyset$, then

$$\varphi(P) = \frac{1}{m} r_1(P|_B) + \frac{1}{m} r_2(P|_B) + \cdots + \frac{1}{m} r_{m-1}(P|_B) + \frac{1}{2m} r_m(P|_B) + \frac{1}{2m} r_{m+1}(P|_B);$$

[37] Part (1) of Proposition 2 and Proposition 1, on which part (1) of Proposition 2 relies, assume strong connectedness. However, one can check that the proofs of these two results only use properties of the environment implied by pairwise connectedness.

[38] Our proof is inspired by an example in Kumar et al. (2021a).

(ii) if $P \in \mathcal{D}(C, B)$ and $\Pi(P^*, P, \mathcal{E}|_{\mathcal{D}(C,B)}) = \emptyset$, then

$$\varphi(P) = \frac{1}{m} r_1(P|_B) + \frac{1}{m} r_2(P|_B) + \cdots + \frac{1}{m} r_m(P|_B); \text{ and}$$

(iii) if $P \notin \mathcal{D}(C, B)$, then

$$\varphi(P) = \frac{1}{m} r_1(P|_B) + \frac{1}{m} r_2(P|_B) + \cdots + \frac{1}{m} r_{m-1}(P|_B) + \frac{1}{2m} r_m(P|_B) + \frac{1}{2m} r_{m+1}(P|_B).$$

Step 2: *Showing that φ is locally strategy-proof but is not strategy-proof.*

First, to show that φ is locally strategy-proof, let $P_1, P_2, P_3 \in \mathcal{D}$ be arbitrary types to which conditions (i), (ii), and (iii) above apply, respectively.[39] By construction, $\text{supp}(\varphi(P_1)) = C \cup \{r_{m+1}(P_1|_B)\}$ and $\text{supp}(\varphi(P_2)) = C$.

Consider P_1. While P_1 has an incentive to report P_2, local misrepresentations cannot give $\varphi(P_2)$ to P_1 because $(P_1, P_2) \notin \mathcal{E}$. Clearly, $\varphi(P_1)\, R_1^{sd}\, \varphi(P_3)$.

Next, consider P_2. By construction, $\varphi(P_2)\, P_2^{sd}\, \varphi(P_1)$ and $\varphi(P_2)\, P_2^{sd}\, \varphi(P_3)$.

Finally, consider P_3. Clearly, $\varphi(P_3)\, R_3^{sd}\, \varphi(P_1)$. It remains to show that $\varphi(P_3)\, P_3^{sd}\, \varphi(P_2)$. Assume that we enumerate elements of B as $\{b_1, b_2, \cdots, b_m, b_{m+1}, \cdots\}$ so that

$$b_1\, P_3\, b_2\, P_3\, \cdots\, P_3\, b_m\, P_3\, b_{m+1}\, P_3\, \cdots.$$

Since $P_3 \notin \mathcal{D}(C, B)$ and $|C| = m$, there is $k \in \{1, \cdots, m\}$ with $b_k \notin C$. This observation, together with $\text{supp}(\varphi(P_2)) = C$, is enough to show that $\varphi(P_3)\, P_3^{sd}\, \varphi(P_2)$.

However, φ is not strategy-proof because $\varphi(P^{**})\, (P^*)^{sd}\, \varphi(P^*)$. Thus, RLGE fails, a contradiction.

A.7 Proof of Theorem 3

To prove Theorem 3, assume that an environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies Property ULLO. We first make preliminary observations on the behavior of locally strategy-proof rules. The following lemma says that if two types have a common upper

[39] Type P_1 may report another type $P_1' \in \mathcal{D}$ to which condition (i) applies. Yet it is clear that $\varphi(P_1)\, R_1^{sd}\, \varphi(P_1')$. A similar comment applies to P_2 and P_3.

contour set, then a locally strategy-proof rule should assign the same probability to that set.

Lemma 1 ("Upper-contour-set (UCS) Lemma"). *Assume that φ is a locally strategy-proof rule. For all $P, P' \in \mathcal{D}$ and each $B \subseteq A$, if B is an upper contour set of both P and P', then $\varphi_B(P) = \varphi_B(P')$.*

Proof. Let $P, P' \in \mathcal{D}$ and $B \subseteq A$ be such that B is an upper contour set of both P and P'. There is a path $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$ satisfying conditions (i) and (ii) of Property ULLO for (P, P', B). Since $B\, P\, (A \backslash B)$ and $B\, P'\, (A \backslash B)$, each $P_\ell \in \{P_1, \cdots, P_k\}$ satisfies $B\, P_\ell\, (A \backslash B)$. By local strategy-proofness, $\varphi_B(P) = \varphi_B(P_1) \geq \varphi_B(P_2) \geq \cdots \geq \varphi_B(P_k) = \varphi_B(P')$. A similar argument involving a path from P' to P shows that $\varphi_B(P) \leq \varphi_B(P')$. Thus, $\varphi_B(P) = \varphi_B(P')$. □

Next, we ask how a locally strategy-proof rule should adjust the probability it assigns to an upper contour set as some alternatives in it become less desirable. Assume that $\{U, V, \{x\}, \{y\}\}$ is a partition of A; and that $P, P' \in \mathcal{D}$ satisfy $P = [U, x, y, V]$, $U\, P'\, (\{y\} \cup V)$, and $(U \cup \{x\})\, P'\, V$. As we go from P to P', x may (or may not) be overtaken by y, so that the upper contour set $U \cup \{x\}$ of P becomes less desirable. Then does a locally strategy-proof rule φ satisfy $\varphi_{U \cup \{x\}}(P) \geq \varphi_{U \cup \{x\}}(P')$ as our intuition suggests? The answer is yes. Further, there are slightly more general cases where probabilities should change in such a direction.

Lemma 2. *Assume that φ is a locally strategy-proof rule. Let $\{U, V, X, Y\}$ be a partition of A, with U and V permitted to be empty. Let $P, P' \in \mathcal{D}$ satisfy $P = [U, X, Y, V]$, $U\, P'\, (Y \cup V)$, $(U \cup X)\, P'\, V$, and $|\{(x, y) \in X \times Y : y\, P'\, x\}| = 1$. Then $\varphi_{U \cup X}(P) \geq \varphi_{U \cup X}(P')$.*

Proof. Choose φ, $\{U, V, X, Y\}$, P, and P' as in the lemma. Write $\{(x, y) \in X \times Y : y\, P'\, x\} = \{(x^*, y^*)\}$. Consider an upper contour set $B \equiv U \cup X$ of P. There is a path $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$ satisfying conditions (i) and (ii) of Property ULLO.[40] By construction, π crosses the x^*-y^* border exactly once.

[40] Condition (ii) of Property ULLO is not needed to prove this lemma.

Let $h \in \{1, \cdots, k-1\}$ be such that edge (P_h, P_{h+1}) crosses the x^*-y^* border. Then $P_h = [U \cup X, Y \cup V]$. By the UCS Lemma, $\varphi_{U \cup X}(P_1) = \varphi_{U \cup X}(P_h)$. Also, since $U \cup X$ is an upper contour set of P_h, local strategy-proofness implies $\varphi_{U \cup X}(P_h) \geq \varphi_{U \cup X}(P_{h+1})$. Finally, since $P_{h+1} = [U \cup (X \setminus \{x^*\}), y^*, x^*, V \cup (Y \setminus \{y^*\})]$ and $P_k = [U \cup (X \setminus \{x^*\}), y^*, x^*, V \cup (Y \setminus \{y^*\})]$, applying the UCS Lemma repeatedly, it follows that $\varphi_{U \cup X}(P_{h+1}) = \varphi_{U \cup X}(P_k)$. In sum, $\varphi_{U \cup X}(P_1) \geq \varphi_{U \cup X}(P_k)$. □

In fact, the above lemma continues to hold when the number of required border crossings for pairs $(x, y) \in X \times Y$ (i.e., $|\{(x, y) \in X \times Y : y \, P' \, x\}|$) is arbitrary.

Lemma 3 ("Overtaken Lemma"). *Assume that φ is a locally strategy-proof rule. Let $\{U, V, X, Y\}$ be a partition of A, with U and V permitted to be empty. Let $P, P' \in \mathcal{D}$ satisfy $P = [U, X, Y, V]$, $U \, P' \, (Y \cup V)$, and $(U \cup X) \, P' \, V$. Then $\varphi_{U \cup X}(P) \geq \varphi_{U \cup X}(P')$.*

Proof. Assume that φ is a locally strategy-proof rule. Let us begin by defining the following statement for each $m \in \mathbb{N}$.

Statement m-Overtaken: Let $\{U, V, X, Y\}$ be a partition of A, with U and V permitted to be empty. Let $P, P' \in \mathcal{D}$ satisfy $P = [U, X, Y, V]$, $U \, P' \, (Y \cup V)$, $(U \cup X) \, P' \, V$, and $|\{(x, y) \in X \times Y : y \, P' \, x\}| \leq m$. Then $\varphi_{U \cup X}(P) \geq \varphi_{U \cup X}(P')$.

We show, by the induction on m, that for each $m \in \mathbb{N}$, Statement m-Overtaken holds. First, Lemma 2 proves Statement m-Overtaken for $m = 1$.[41]

Next, assume the following **Induction Hypothesis:** for some $m \in \mathbb{N}$, Statement m-Overtaken holds. We show that Statement $(m+1)$-Overtaken holds. Let $\{U, V, X, Y\}$ be a partition of A, with U and V permitted to be empty. Let $P, P' \in \mathcal{D}$ satisfy $P = [U, X, Y, V]$, $U \, P' \, (Y \cup V)$, $(U \cup X) \, P' \, V$, and $|\Omega| \leq m + 1$, where $\Omega \equiv \{(x, y) \in X \times Y : y \, P' \, x\}$. To avoid triviality, assume that $|\Omega| = m + 1$. Take an upper contour set $B \equiv U \cup X$ of P. By Property ULLO, there is a path $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$ such that (i) for each $x \in B$ and each

[41] When $m = 0$, statement m-Overtaken trivially follows from the UCS Lemma.

$y \in L(P,B) = A \backslash B$, π is without $\{x,y\}$-restoration; and (ii) for each $\ell \in \{1,\cdots,k-1\}$, letting $\{C_1^\ell,\cdots,C_{t_\ell}^\ell\}$ be the associated partition for $(P_\ell, P_{\ell+1})$, for each $s \in \{1,\cdots,t_\ell\}$, $P_\ell|_{C_s^\ell} = [C_s^\ell \cap B, C_s^\ell \backslash B]$ or $P_{\ell+1}|_{C_s^\ell} = [C_s^\ell \backslash B, C_s^\ell \cap B]$. For each $(x,y) \in \Omega$, $x\, P\, y$ and $y\, P'\, x$, so that π crosses the x-y border exactly once. We distinguish two cases.

Case 1: *There is $h \in \{1,\cdots,k-1\}$ such that for each $(x,y) \in \Omega$, edge (P_h, P_{h+1}) crosses the x-y border.*

For any $(x,y) \in \Omega$, the subpath $\pi|_{[P_1,P_h]}$ does not cross the x-y border. Thus, $P_h = [U \cup X, Y \cup V]$. By the UCS Lemma, $\varphi_{U\cup X}(P_1) = \varphi_{U\cup X}(P_h)$. Also, since $U \cup X$ is an upper contour set of P_h, local strategy-proofness implies $\varphi_{U\cup X}(P_h) \geq \varphi_{U\cup X}(P_{h+1})$.

Finally, we show that $\varphi_{U\cup X}(P_{h+1}) = \varphi_{U\cup X}(P_k)$. Note that we may write $P_{h+1} = [U \cup X_1, Y_1, X_2, Y_2, \cdots, X_r, Y_r \cup V]$, where $\{X_1,\cdots,X_r\}$ and $\{Y_1,\cdots,Y_r\}$ are partitions of X and Y, respectively, with X_1 and Y_r permitted to be empty. By our assumption on h, for any $(x,y) \in \Omega$, the subpath $\pi|_{[P_{h+1},P_k]}$ does not cross the x-y border. This means that $P_k = [U \cup X_1, Y_1, X_2, Y_2, \cdots, X_r, Y_r \cup V]$. Applying the UCS Lemma repeatedly, it follows that $\varphi_{U\cup X_1}(P_{h+1}) = \varphi_{U\cup X_1}(P_k)$, $\varphi_{X_2}(P_{h+1}) = \varphi_{X_2}(P_k), \cdots, \varphi_{X_r}(P_{h+1}) = \varphi_{X_r}(P_k)$, so that $\varphi_{U\cup X}(P_{h+1}) = \varphi_{U\cup X}(P_k)$.

Case 2: *Case 1 does not hold.*

Let $h \in \{1,\cdots,k-1\}$ be such that (P_h, P_{h+1}) is the last edge in π that crosses the x'-y' border for some $(x',y') \in \Omega$. We proceed in three steps.

Step 1: *Showing that $\varphi_B(P_1) \geq \varphi_B(P_h)$.*

Consider P_1 and P_h. Observe that $P_1 = P = [U,X,Y,V]$, $U\, P_h\, (Y \cup V)$, $(U \cup X)\, P_h\, V$, and $|\{(x,y) \in X \times Y : y\, P_h\, x\}| \leq m$.[42] By the Induction Hypothesis applied to P_1 and P_h, $\varphi_B(P_1) \geq \varphi_B(P_h)$.

Step 2: *Showing that $\varphi_B(P_h) \geq \varphi_B(P_{h+1})$.*

Consider P_h, P_{h+1}, and the associated partition $\{C_1^h,\cdots,C_{t_h}^h\}$ for edge (P_h, P_{h+1}). Observe that $\varphi_B(P_h) = \sum_{s\in\{1,\cdots,t_h\}} \varphi_{C_s^h \cap B}(P_h)$ and $\varphi_B(P_{h+1}) = \sum_{s\in\{1,\cdots,t_h\}} \varphi_{C_s^h \cap B}(P_{h+1})$.

[42] To see the inequality, let $\Omega_h \equiv \{(x,y) \in X \times Y : y\, P_h\, x\}$. First, $\Omega_h \subseteq \Omega$ because if there is $(x,y) \in \Omega_h \backslash \Omega$, then $x\, P_1\, y$, $y\, P_h\, x$, and $x\, P_k\, y$, so that π is with $\{x,y\}$-restoration, a contradiction. Further, in fact, $\Omega_h \subsetneq \Omega$ because $(x',y') \in \Omega \backslash \Omega_h$.

To show that $\varphi_B(P_h) \geq \varphi_B(P_{h+1})$, we show that for each $s \in \{1, \cdots, t_h\}$, $\varphi_{C_s^h \cap B}(P_h) \geq \varphi_{C_s^h \cap B}(P_{h+1})$. Let $s \in \{1, \cdots, t_h\}$. Let us distinguish two cases.

First, assume that $P_h|_{C_s^h} = [C_s^h \cap B, C_s^h \backslash B]$. Since $C_1^h \cup \cdots \cup C_{s-1}^h \cup (C_s^h \cap B)$ is an upper contour set of P_h, local strategy-proofness implies $\varphi_{C_1^h \cup \cdots \cup C_{s-1}^h \cup (C_s^h \cap B)}(P_h) \geq \varphi_{C_1^h \cup \cdots \cup C_{s-1}^h \cup (C_s^h \cap B)}(P_{h+1})$. By the UCS Lemma, $\varphi_{C_1^h \cup \cdots \cup C_{s-1}^h}(P_h) = \varphi_{C_1^h \cup \cdots \cup C_{s-1}^h}(P_{h+1})$. Thus, $\varphi_{C_s^h \cap B}(P_h) \geq \varphi_{C_s^h \cap B}(P_{h+1})$.

Second, assume that $P_{h+1}|_{C_s^h} = [C_s^h \backslash B, C_s^h \cap B]$. Define $\hat{U} \equiv C_1^h \cup \cdots \cup C_{s-1}^h$, $\hat{V} \equiv C_{s+1}^h \cup \cdots \cup C_{t_h}^h$, $\hat{X} \equiv C_s^h \backslash B$, and $\hat{Y} \equiv C_s^h \cap B$. Using this notation, we have $P_{h+1} = [\hat{U}, \hat{X}, \hat{Y}, \hat{V}]$, $\hat{U}\, P_h\, (\hat{Y} \cup \hat{V})$, $(\hat{U} \cup \hat{X})\, P_h\, \hat{V}$, and $|\{(x, y) \in \hat{X} \times \hat{Y} : y\, P_h\, x\}| \leq m$.[43] Thus, by the Induction Hypothesis applied to P_{h+1} and P_h, $\varphi_{\hat{U} \cup \hat{X}}(P_{h+1}) \geq \varphi_{\hat{U} \cup \hat{X}}(P_h)$. Since $P_{h+1} = [\hat{U}, \hat{X} \cup \hat{Y}, \hat{V}]$ and $P_h = [\hat{U}, \hat{X} \cup \hat{Y}, \hat{V}]$, the UCS Lemma implies $\varphi_{\hat{U}}(P_{h+1}) = \varphi_{\hat{U}}(P_h)$ and $\varphi_{\hat{X} \cup \hat{Y}}(P_{h+1}) = \varphi_{\hat{X} \cup \hat{Y}}(P_h)$. Thus, $\varphi_{\hat{Y}}(P_{h+1}) \leq \varphi_{\hat{Y}}(P_h)$, i.e., $\varphi_{C_s^h \cap B}(P_{h+1}) \leq \varphi_{C_s^h \cap B}(P_h)$.

Step 3: *Showing that* $\varphi_B(P_{h+1}) = \varphi_B(P_k)$.

This follows from an argument similar to that in the last paragraph of Case 1.

□

With the aid of the lemmas above, we now prove Theorem 3. Assume that an environment $\langle \mathcal{D}, \mathcal{E} \rangle$ satisfies Property ULLO. Let φ be a locally strategy-proof rule. To show that it is strategy-proof, let $P, P' \in \mathcal{D}$. Let $B \subseteq A$ be an upper contour set of P. By Property ULLO, there is a path $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$ such that (i) for each $b \in B$ and each $c \in L(P, B) = A \backslash B$, π is without $\{b, c\}$-restoration; and (ii) for each $\ell \in \{1, \cdots, k-1\}$, letting $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ be the associated partition for $(P_\ell, P_{\ell+1})$, for each $s \in \{1, \cdots, t_\ell\}$, $P_\ell|_{C_s^\ell} = [C_s^\ell \cap B, C_s^\ell \backslash B]$ or $P_{\ell+1}|_{C_s^\ell} = [C_s^\ell \backslash B, C_s^\ell \cap B]$.

We prove $\varphi_B(P_1) \geq \varphi_B(P_k)$ by induction. First, $\varphi_B(P_1) \geq \varphi_B(P_2)$ since $(P_1, P_2) \in \mathcal{E}$ and φ is locally strategy-proof.

[43] To see the inequality, let $\hat{\Omega} \equiv \{(x, y) \in \hat{X} \times \hat{Y} : y\, P_h\, x\}$. First, we show that for each $(x, y) \in \hat{\Omega}$, $(y, x) \in \Omega$, so that $|\hat{\Omega}| \leq |\Omega|$. Let $(x, y) \in \hat{\Omega}$. First, $x\, P_{h+1}\, y$. Also, $x \in \hat{X} \subseteq A \backslash B$ and $y \in \hat{Y} \subseteq B$, so that $y\, P\, x$. Since π is without $\{x, y\}$-restoration, $x\, P'\, y$. Since $U\, P'\, (Y \cup V)$, $(U \cup X)\, P'\, V$, $y \notin U$ and $x \notin V$, so that $y \in X$ and $x \in Y$, and hence $(y, x) \in \Omega$.

By the assumption for Case 2, there is some $(y^*, x^*) \in \Omega$ such that π crosses the y^*-x^* border at one of the edges $(P_1, P_2), \cdots, (P_{h-1}, P_h)$. Thus, $x^*\, P_h\, y^*$ and (since π is without $\{x^*, y^*\}$-restoration) $x^*\, P_{h+1}\, y^*$. Then $(x^*, y^*) \notin \hat{\Omega}$. Thus, $|\hat{\Omega}| < |\Omega| \leq m + 1$.

Next, let us assume the following **Induction Hypothesis:** for some $\ell \in \{1, \cdots, k-1\}$, $\varphi_B(P_1) \geq \varphi_B(P_\ell)$. We now show $\varphi_B(P_1) \geq \varphi_B(P_{\ell+1})$ by showing $\varphi_B(P_\ell) \geq \varphi_B(P_{\ell+1})$. Consider the associated partition $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ of A for $(P_\ell, P_{\ell+1})$. Observe that $\varphi_B(P_\ell) = \sum_{s \in \{1,\cdots,t_\ell\}} \varphi_{C_s^\ell \cap B}(P_\ell)$ and $\varphi_B(P_{\ell+1}) = \sum_{s \in \{1,\cdots,t_\ell\}} \varphi_{C_s^\ell \cap B}(P_{\ell+1})$. To show $\varphi_B(P_\ell) \geq \varphi_B(P_{\ell+1})$, we show that for each $s \in \{1, \cdots, t_\ell\}$, $\varphi_{C_s^\ell \cap B}(P_\ell) \geq \varphi_{C_s^\ell \cap B}(P_{\ell+1})$. Let $s \in \{1, \cdots, t_\ell\}$. We distinguish two cases.

Case 1: $P_\ell|_{C_s^\ell} = [C_s^\ell \cap B, C_s^\ell \backslash B]$.

Define $U \equiv C_1^\ell \cup \cdots \cup C_{s-1}^\ell$, $V \equiv C_{s+1}^\ell \cup \cdots \cup C_{t_\ell}^\ell$, $X \equiv C_s^\ell \cap B$, and $Y \equiv C_s^\ell \backslash B$. With this notation, we have $P_\ell = [U, X, Y, V]$ and $P_{\ell+1} = [U, X \cup Y, V]$. Since $(P_\ell, P_{\ell+1}) \in \mathcal{E}$, local strategy-proofness implies $\varphi_{U \cup X}(P_\ell) \geq \varphi_{U \cup X}(P_{\ell+1})$. Also, by the UCS Lemma, $\varphi_U(P_\ell) = \varphi_U(P_{\ell+1})$. Thus, $\varphi_X(P_\ell) \geq \varphi_X(P_{\ell+1})$, i.e., $\varphi_{C_s^\ell \cap B}(P_\ell) \geq \varphi_{C_s^\ell \cap B}(P_{\ell+1})$.

Case 2: $P_{\ell+1}|_{C_s^\ell} = [C_s^\ell \backslash B, C_s^\ell \cap B]$.

Define $U \equiv C_1^\ell \cup \cdots \cup C_{s-1}^\ell$, $V \equiv C_{s+1}^\ell \cup \cdots \cup C_{t_\ell}^\ell$, $X \equiv C_s^\ell \backslash B$, and $Y \equiv C_s^\ell \cap B$. With this notation, we have $P_{\ell+1} = [U, X, Y, V]$, $U \, P_\ell \, (Y \cup V)$, and $(U \cup X) \, P_\ell \, V$. Thus, by the Overtaken Lemma applied to $P_{\ell+1}$ and P_ℓ, $\varphi_{U \cup X}(P_{\ell+1}) \geq \varphi_{U \cup X}(P_\ell)$. Since $P_{\ell+1} = [U, X \cup Y, V]$ and $P_\ell = [U, X \cup Y, V]$, the UCS Lemma implies $\varphi_U(P_{\ell+1}) = \varphi_U(P_\ell)$ and $\varphi_{X \cup Y}(P_{\ell+1}) = \varphi_{X \cup Y}(P_\ell)$. Thus, $\varphi_Y(P_{\ell+1}) \leq \varphi_Y(P_\ell)$, i.e., $\varphi_{C_s^\ell \cap B}(P_{\ell+1}) \leq \varphi_{C_s^\ell \cap B}(P_\ell)$.

A.8 Proof of Proposition 5

Let $\langle \mathcal{D}, \mathcal{E} \rangle$ be an environment satisfying ULGO. Let $P, P' \in \mathcal{D}$. Let $B \subseteq A$ be an upper contour set of P. There is a path $\pi \equiv (P_1, \cdots, P_k) \in \Pi(P, P')$ such that (i) for each $b \in B$ and each $c \in L(P, B) = A \backslash B$, π is without $\{b, c\}$-restoration; and (ii) for each $\ell \in \{1, \cdots, k-1\}$, letting $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ be the associated partition for $(P_\ell, P_{\ell+1})$, for each $s \in \{1, \cdots, t_\ell\}$, $|C_s^\ell| \leq 3$. Let $\ell \in \{1, \cdots, k-1\}$ and $s \in \{1, \cdots, t_\ell\}$. Consider C_s^ℓ. The proof will be complete if we show that $P_\ell|_{C_s^\ell} = [C_s^\ell \cap B, C_s^\ell \backslash B]$ or $P_{\ell+1}|_{C_s^\ell} = [C_s^\ell \backslash B, C_s^\ell \cap B]$. If $|C_s^\ell \cap B| = 0$ or $|C_s^\ell \cap B| = 3$, then the claim is trivial. Thus, we only consider the following

two cases.

Case 1: $|C_s^\ell \cap B| = 1$.

If $|C_s^\ell| = 1$, then the claim is trivial again. Thus, we only consider the following two cases.

Case 1.1: $|C_s^\ell| = 2$.

Let us write $C_s^\ell = \{a, b\}$ and $C_s^\ell \cap B = \{a\}$. Note that $a \, P \, b$. Since P_ℓ and $P_{\ell+1}$ have the opposite preferences over $\{a, b\}$, π is without $\{a, b\}$-restoration only if $a \, P_\ell \, b$ and $b \, P_{\ell+1} \, a$. Then $P_\ell|_{C_s^\ell} = [C_s^\ell \cap B, C_s^\ell \backslash B]$ (and $P_{\ell+1}|_{C_s^\ell} = [C_s^\ell \backslash B, C_s^\ell \cap B]$).

Case 1.2: $|C_s^\ell| = 3$.

Let us write $C_s^\ell = \{a, b, c\}$ and $C_s^\ell \cap B = \{a\}$. Note that $a \, P \, \{b, c\}$.

First, we show that $a \neq r_3(P_\ell|_{C_s^\ell})$. Suppose, by contradiction, that $a = r_3(P_\ell|_{C_s^\ell})$, i.e., $P_\ell|_{C_s^\ell} = [\{b, c\}, a]$. Since $a \, P \, \{b, c\}$, $\{b, c\} \, P_\ell \, a$, and π is without $\{a, b\}$- and $\{a, c\}$-restoration, $\{b, c\} \, P_{\ell+1} \, a$. That is, $P_{\ell+1}|_{C_s^\ell} = [\{b, c\}, a]$, which contradicts that $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ is the finest associated partition for edge $(P_\ell, P_{\ell+1})$.

If $a = r_1(P_\ell|_{C_s^\ell})$, then clearly, $P_\ell|_{C_s^\ell} = [a, \{b, c\}] = [C_s^\ell \cap B, C_s^\ell \backslash B]$.

Next, assume that $a = r_2(P_\ell|_{C_s^\ell})$, say, (without loss of generality) $P_\ell|_{C_s^\ell} = [b, a, c]$. Since $a \, P \, b$, $b \, P_\ell \, a$, and π is without $\{a, b\}$-restoration, $P_{\ell+1}$ satisfies $b \, P_{\ell+1} \, a$. Further, since $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ is the finest associated partition for edge $(P_\ell, P_{\ell+1})$, $P_{\ell+1}|_{C_s^\ell} \neq [b, a, c], [b, c, a]$. Thus, $P_{\ell+1}|_{C_s^\ell} = [c, b, a] = [C_s^\ell \backslash B, C_s^\ell \cap B]$.

Case 2: $|C_s^\ell \cap B| = 2$.

If $|C_s^\ell| = 2$, then the claim is trivial. Thus, assume that $|C_s^\ell| = 3$. Let us write $C_s^\ell = \{a, b, c\}$ and $C_s^\ell \cap B = \{a, b\}$. Note that $\{a, b\} \, P \, c$.

First, we show that $r_1(P_\ell|_{C_s^\ell}) \neq c$. Suppose, by contradiction, that $r_1(P_\ell|_{C_s^\ell}) = c$. Since $\{a, b\} \, P \, c$, $c \, P_\ell \, \{a, b\}$, and π is without $\{a, c\}$- and $\{b, c\}$-restoration, $P_{\ell+1}$ satisfies $c \, P_{\ell+1} \, \{a, b\}$. This, however, contradicts that $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ is the finest associated partition for edge $(P_\ell, P_{\ell+1})$.

By the claim in the previous paragraph, $r_1(P_\ell|_{C_s^\ell}) = a$ or b. Without loss of generality, assume that $r_1(P_\ell|_{C_s^\ell}) = a$. If $P_\ell|_{C_s^\ell} = [a, b, c]$, then $P_\ell|_{C_s^\ell} = [C_s^\ell \cap B, C_s^\ell \backslash B]$. Thus, assume, henceforth, that $P_\ell|_{C_s^\ell} = [a, c, b]$. Since $b \, P \, c$, $c \, P_\ell \, b$, and

π is without $\{b, c\}$-restoration, $c\, P_{\ell+1}\, b$, so that $P_{\ell+1}|_{C_s^\ell} \neq [a, b, c], [b, a, c], [b, c, a]$. Also, since $\{C_1^\ell, \cdots, C_{t_\ell}^\ell\}$ is the finest associated partition for edge $(P_\ell, P_{\ell+1})$, $P_{\ell+1}|_{C_s^\ell} \neq [a, b, c], [a, c, b], [c, a, b]$. The only remaining possibility is $P_{\ell+1}|_{C_s^\ell} = [c, b, a] = [C_s^\ell \backslash B, C_s^\ell \cap B]$.

References

[1] Allingham, M., and A. Sandmo, 1972, "Income tax evasion: A theoretical analysis", *Journal of Public Economics*, 1, 323–338.

[2] Altuntaş, A., W. Phan, and Y. Tamura, 2023, "Some characterizations of generalized top trading cycles", *Games and Economic Behavior*, 141, 156–181.

[3] Andreoni, J., B. Erard, and J. Feinstein, 1998, "Tax compliance", *Journal of Economic Literature*, 36, 818–860.

[4] Archer, A., and R. Kleinberg, 2014, "Truthful germs are contagious: A local-to-global characterization of truthfulness", *Games and Economic Behavior*, 86, 340–366.

[5] Black, D., 1958, *The Theory of Committees and Elections*, Cambridge University Press, London.

[6] Bogomolnaia, A., and H. Moulin, 2001. "A new solution to the random assignment problem", *Journal of Economic Theory*, 100, 295–328.

[7] Bull, J., and J. Watson, 2007, "Hard evidence and mechanism design", *Games and Economic Behavior*, 58, 75–93.

[8] Can, B., and T. Storcken, 2018, "A re-characterization of the Kemeny distance", *Journal of Mathematical Economics*, 79, 112–116.

[9] Carroll, G., 2012, "When are local incentive constraints sufficient?", *Econometrica*, 80, 661–686.

[10] Celik, G., 2006, "Mechanism design with weaker incentive compatibility constraints", *Games and Economic Behavior*, 56, 37–44.

[11] Charness, G., and M. Dufwenberg, 2006, "Promises and partnership", *Econometrica*, 74, 1579–1601.

[12] Chatterji, S., S. Roy, S. Sadhukhan, A. Sen, and H. Zeng, 2022, "Probabilistic fixed ballot rules and hybrid domains", *Journal of Mathematical Economics*, 100, 102656.

[13] Chatterji, S., and H. Zeng, 2018, "On random social choice functions with the tops-only property", *Games Economic Behavior*, 109, 413–435.

[14] Che, Y.-K., and I. Gale, 2000, "The optimal mechanism for selling to a budget-constrained buyer", *Journal of Economic Theory*, 92, 198–233.

[15] Cho, W. J., 2016, "Incentive properties for ordinal mechanisms", *Games and Economic Behavior*, 95, 168–177.

[16] Cho, W. J., and C. Park, 2025, "The local-global equivalence on general networks: applications", working paper, available at SSRN(https://ssrn.com/abstract=5219192).

[17] d'Aspremont, C., and B. Peleg, 1988, "Ordinal Bayesian incentive compatible representations of committees", *Social Choice and Welfare*, 5, 261–279.

[18] Deneckere, R., and S. Severinov, 2008, "Mechanism design with partial state verifiability", *Games and Economic Behavior*, 64, 487–513.

[19] Dufwenberg, M., and M. A. Dufwenberg, 2018, "Lies in disguise–A theoretical analysis of cheating", *Journal of Economic Theory*, 175, 248–264.

[20] Fischbacher, U., and F. Föllmi-Heusi, 2013, "Lies in disguise–An experimental study on cheating", *Journal of the European Economic Association*, 11, 525–547.

[21] Forges, F., and F. Koessler, 2005, "Communication equilibria with partially verifiable types", *Journal of Mathematical Economics*, 41, 793–811.

[22] Gibbard, A., 1973, "Manipulation of voting schemes: A general result", *Econometrica*, 41, 587–601.

[23] Gibbard, A., 1977, "Manipulation of schemes that mix voting with chance", *Econometrica*, 45, 665–681.

[24] Gneezy, U., 2005, "Deception: The role of consequence", *American Economic Review*, 95, 384–394.

[25] Green, J. R., and J.-J. Laffont, 1986, "Partially verifiable information and mechanism design", *Review of Economic Studies*, 53, 3, 447–456.

[26] Grossman, S. J., 1981, "The informational role of warranties and private disclosure about product quality", *Journal of Law and Economics*, 24, 461–483.

[27] Hong, M., and S. Kim, 2023, "Unanimity and local incentive compatibility in sparsely connected domains", *Social Choice and Welfare*, https://doi.org/10.1007/s00355-023-01457-3.

[28] Hurkens, S., and N. Kartik, 2009, "Would I lie to you? On social preferences and lying aversion", *Experimental Economics*, 12, 180–192.

[29] Inada, K., 1964. "A note on the simple majority decision rule", *Econometrica*, 32, 525–531.

[30] Karmokar, M., and S. Roy, 2022, "The structure of (local) ordinal Bayesian incentive compatible random rules", *Economic Theory*, 76, 111–152.

[31] Kartik, N., 2009, "Strategic communication with lying costs", *Review of Economic Studies*, 76, 1359-1395.

[32] Kemeny, J., 1959, "Mathematics without numbers", *Daedalus*, 88, 577–591.

[33] Krähmer, D., and R. Strausz, 2024, "Unidirectional incentive compatibility", mimeo.

[34] Kumar, U., S. Roy, A. Sen, S. Yadav, and H. Zeng, 2021a, "Local-global equivalence in voting models: A characterization and applications", *Theoretical Economics*, 16, 1195–1220.

[35] Kumar, U., S. Roy, A. Sen, S. Yadav, and H. Zeng, 2021b, "Local global equivalence for unanimous social choice functions", *Games and Economic Behavior*, 130, 299–308.

[36] Kumar, U., S. Roy, A. Sen, S. Yadav, and H. Zeng, 2023, "Directed graph LGE", private communication.

[37] Maggi, G., and A. Rodríguez-Clare, 1995, "Costly distortion of information in agency problems", *The RAND Journal of Economics*, 26, 675–689.

[38] Mazar, N., O. Amir, and D. Ariely, 2008, "The dishonesty of honest people: A theory of self-concept maintenance", *Journal of Marketing Research*, 45, 633–644.

[39] Milgrom, P. R., 1981, "Good news and bad news: Representation theorems and applications", *The Bell Journal of Economics*, 12, 380–391.

[40] Milgrom, P. R., and J. Roberts, 1986, "Relying on the information of interested parties", *The RAND Journal of Economics*, 17, 1, 18–32.

[41] Mishra, D., 2016, "Ordinal Bayesian incentive compatibility in restricted domains", *Journal of Economic Theory*, 163, 925–954.

[42] Mishra, D., A. Pramanik, and S. Roy, 2016, "Local incentive compatibility with transfers", *Games and Economic Behavior*, 100, 149–165.

[43] Nagin, D., J. Rebitzer, S. Sanders, and L. Taylor, 2002, "Monitoring, motivation, and management: The determinants of opportunistic behavior in a field experiment", *American Economic Review*, 92, 850–873.

[44] Okuno-Fujiwara, M., A. Postlewaite, and K. Suzumura, 1990, "Strategic information revelation", *Review of Economic Studies*, 57, 25–47.

[45] Roy, S., and S. Sadhukhan, 2022, "On the equivalence of strategy-proofness and upper contour strategy-proofness for randomized social choice functions", *Journal of Mathematical Economics*, 99, 102593.

[46] Sánchez-Pagés, S., and M. Vorsatz, 2007, "An experimental study of truth-telling in a sender-receiver game", *Games and Economic Behavior*, 61, 86–112.

[47] Sato, S., 2013a, "A sufficient condition for the equivalence of strategy-proofness and nonmanipulability by preferences adjacent to the sincere one", *Journal of Economic Theory*, 148, 259–278.

[48] Sato, S., 2013b, "Strategy-proofness and the reluctance to make large lies: The case of weak orders", *Social Choice and Welfare*, 40, 479–494.

[49] Satterthwaite, M., 1975, "Strategy-proofness and Arrow's conditions: Existence and correspondence theorems for voting procedures and social welfare functions", *Journal of Economic Theory*, 10, 187–217.

[50] Sen, A. K., 1966, "A possibility theorem on majority decisions", *Econometrica*, 34, 491–499.

[51] Sher, I., and R. Vohra, 2015, "Price discrimination through communication", *Theoretical Economics*, 10, 597–648.

[52] Torgler, B., 2002, "Speaking to theorists and searching for facts: Tax morale and tax compliance in experiments", *Journal of Economic Surveys*, 16, 657–683.

[53] Vanberg, C., 2008, "Why do people keep their promises? An experimental test of two explanations", *Econometrica*, 76, 1467–1480.

[54] Vickrey, W., 1960. "Utility, strategy, and social decision rules", *Quarterly Journal of Economics*, 75, 507–535.